SampleSize_n100_5Treatments/table2_nonshrink_5_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.66	89.66	89.66	93.10	100.00	82.76	96.55	93.1	96.55	96.55	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	96.43	89.29	100.00	89.29	96.43	100.0	96.43	92.86	100.00
20%	100.00	100.00	96.43	89.29	96.43	85.71	100.00	100.0	96.43	96.43	100.00
30%	93.10	93.10	96.55	89.66	89.66	75.86	100.00	100.0	96.55	96.55	96.55
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.0	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	92.86	100.00	89.29	96.43	100.0	96.43	92.86	100.00
20%	96.43	96.43	92.86	89.29	92.86	78.57	100.00	100.0	96.43	96.43	100.00
30%	93.10	93.10	96.55	93.10	86.21	75.86	100.00	100.0	96.55	100.00	100.00
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.0	96.55	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	96.43	92.86	100.00	89.29	96.43	100.0	96.43	92.86	100.00
20%	100.00	96.43	92.86	89.29	92.86	78.57	100.00	100.0	96.43	96.43	100.00
30%	93.10	89.66	89.66	89.66	86.21	72.41	100.00	100.0	100.00	93.10	93.10
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.0	100.00	100.00	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	89.66	89.66	93.10	93.10	72.41	100.00	100.0	96.55	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_5Treatments/table2_nonshrink_5_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.43	96.43	92.86	96.43	96.43	82.14	100.00	100.00	100.00	100.00	96.43
20%	96.55	96.55	96.55	93.10	100.00	89.66	100.00	100.00	100.00	96.55	100.00
30%	100.00	89.66	96.55	93.10	93.10	79.31	100.00	93.10	96.55	96.55	96.55
40%	96.43	96.43	100.00	96.43	92.86	89.29	96.43	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	100.00	96.43	82.14	100.00	100.00	100.00	100.00	100.00
20%	93.10	96.55	96.55	93.10	100.00	86.21	100.00	100.00	100.00	96.55	100.00
30%	100.00	86.21	96.55	89.66	93.10	75.86	100.00	93.10	96.55	96.55	100.00
40%	96.43	92.86	96.43	96.43	89.29	78.57	96.43	100.00	100.00	100.00	92.86
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	92.86	96.43	82.14	100.00	100.00	100.00	100.00	100.00
20%	96.55	96.55	96.55	93.10	96.55	89.66	100.00	96.55	100.00	96.55	100.00
30%	96.55	86.21	96.55	89.66	93.10	72.41	100.00	93.10	96.55	96.55	100.00
40%	96.43	92.86	96.43	96.43	92.86	82.14	96.43	100.00	100.00	100.00	96.43

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	89.66
Imputation using linear regression	NA
10%	89.29
20%	92.86
30%	93.10
40%	100.00
Imputation using predictive mean matching method	NA
10%	89.29
20%	92.86
30%	96.55
40%	96.55
Imputation using propensity−score method	NA
10%	89.29
20%	92.86
30%	86.21
40%	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	96.43
20%	96.55
30%	89.66
40%	96.43
Imputation using predictive mean matching method	NA
10%	100.00
20%	96.55
30%	93.10
40%	92.86
Imputation using propensity−score method	NA
10%	100.00
20%	93.10
30%	93.10
40%	92.86

SampleSize_n100_5Treatments/table2_shrink_5_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.66	89.66	89.66	93.10	100.00	82.76	96.55	96.55	96.55	96.55	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	96.43	89.29	100.00	89.29	96.43	100.00	96.43	96.43	100.00
20%	100.00	100.00	96.43	89.29	96.43	85.71	100.00	100.00	96.43	96.43	100.00
30%	93.10	93.10	96.55	89.66	89.66	75.86	100.00	100.00	96.55	96.55	96.55
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	92.86	100.00	89.29	96.43	100.00	96.43	92.86	100.00
20%	96.43	96.43	92.86	89.29	92.86	78.57	100.00	100.00	96.43	96.43	100.00
30%	93.10	93.10	96.55	93.10	86.21	75.86	100.00	100.00	96.55	100.00	93.10
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.00	96.55	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	96.43	92.86	100.00	89.29	96.43	100.00	96.43	96.43	100.00
20%	100.00	96.43	92.86	89.29	92.86	78.57	100.00	100.00	96.43	96.43	100.00
30%	93.10	89.66	89.66	89.66	86.21	72.41	100.00	100.00	100.00	93.10	93.10
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.00	100.00	100.00	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	89.66	89.66	93.10	93.10	72.41	100.00	100.00	96.55	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_5Treatments/table2_shrink_5_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.43	96.43	92.86	96.43	96.43	82.14	100.00	100.00	100.00	100.00	96.43
20%	96.55	96.55	96.55	93.10	100.00	89.66	100.00	100.00	100.00	96.55	100.00
30%	100.00	89.66	96.55	93.10	93.10	79.31	100.00	93.10	96.55	96.55	96.55
40%	96.43	96.43	100.00	96.43	92.86	89.29	96.43	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	100.00	96.43	82.14	100.00	100.00	100.00	100.00	100.00
20%	93.10	96.55	96.55	93.10	100.00	86.21	100.00	100.00	100.00	96.55	100.00
30%	100.00	86.21	96.55	89.66	93.10	75.86	100.00	93.10	96.55	96.55	100.00
40%	96.43	92.86	96.43	96.43	89.29	78.57	96.43	100.00	100.00	100.00	92.86
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	92.86	96.43	82.14	100.00	100.00	100.00	100.00	96.43
20%	96.55	96.55	96.55	93.10	96.55	89.66	100.00	96.55	100.00	96.55	100.00
30%	96.55	86.21	96.55	89.66	93.10	72.41	100.00	93.10	96.55	93.10	100.00
40%	96.43	92.86	96.43	96.43	92.86	82.14	96.43	100.00	100.00	100.00	96.43

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	89.66
Imputation using linear regression	NA
10%	92.86
20%	92.86
30%	93.10
40%	100.00
Imputation using predictive mean matching method	NA
10%	89.29
20%	92.86
30%	93.10
40%	96.55
Imputation using propensity−score method	NA
10%	92.86
20%	92.86
30%	86.21
40%	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	96.43
20%	96.55
30%	89.66
40%	96.43
Imputation using predictive mean matching method	NA
10%	100.00
20%	96.55
30%	93.10
40%	92.86
Imputation using propensity−score method	NA
10%	96.43
20%	93.10
30%	89.66
40%	92.86

SampleSize_n100_5Treatments/table3_nonshrink_5_treatments_MAR_n100

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.753	1.766	1.793	1.781	1.799	2.235	2.252	2.286	2.271	2.295
Imputation using linear regression										
10%	1.838 (4.87)	1.842 (4.30)	1.897 (5.80)	1.875 (5.31)	1.901 (5.63)	2.348 (5.03)	2.352 (4.45)	2.423 (5.96)	2.395 (5.46)	2.427 (5.79)
20%	1.928 (10.02)	1.944 (10.10)	1.978 (10.35)	1.981 (11.23)	1.988 (10.50)	2.465 (10.31)	2.486 (10.38)	2.529 (10.63)	2.532 (11.51)	2.543 (10.81)
30%	2.074 (18.33)	2.120 (20.04)	2.147 (19.77)	2.112 (18.63)	2.206 (22.63)	2.645 (18.35)	2.705 (20.09)	2.738 (19.77)	2.695 (18.68)	2.815 (22.67)
40%	2.281 (30.18)	2.262 (28.10)	2.327 (29.83)	2.312 (29.87)	2.372 (31.86)	2.916 (30.44)	2.892 (28.39)	2.975 (30.12)	2.957 (30.21)	3.033 (32.16)
Imputation using predictive mean matching method										
10%	1.824 (4.10)	1.835 (3.92)	1.882 (4.96)	1.863 (4.62)	1.897 (5.44)	2.328 (4.17)	2.342 (3.98)	2.401 (5.03)	2.377 (4.69)	2.421 (5.50)
20%	1.875 (7.01)	1.898 (7.48)	1.929 (7.60)	1.920 (7.80)	1.966 (9.26)	2.400 (7.36)	2.428 (7.82)	2.468 (7.94)	2.456 (8.15)	2.515 (9.60)
30%	1.958 (11.71)	2.039 (15.44)	2.073 (15.65)	2.017 (13.27)	2.088 (16.07)	2.499 (11.80)	2.601 (15.51)	2.646 (15.71)	2.575 (13.37)	2.666 (16.18)
40%	2.152 (22.77)	2.165 (22.62)	2.231 (24.43)	2.192 (23.09)	2.261 (25.66)	2.742 (22.67)	2.758 (22.46)	2.842 (24.28)	2.793 (23.00)	2.881 (25.55)
Imputation using propensity−score method										
10%	1.830 (4.42)	1.836 (3.99)	1.888 (5.32)	1.857 (4.31)	1.895 (5.32)	2.337 (4.55)	2.345 (4.11)	2.411 (5.44)	2.372 (4.43)	2.419 (5.44)
20%	1.892 (7.97)	1.920 (8.72)	1.952 (8.91)	1.934 (8.63)	1.983 (10.23)	2.420 (8.27)	2.455 (9.01)	2.496 (9.19)	2.474 (8.93)	2.536 (10.53)
30%	1.987 (13.36)	2.052 (16.19)	2.048 (14.22)	2.049 (15.09)	2.105 (17.00)	2.536 (13.46)	2.619 (16.29)	2.614 (14.33)	2.616 (15.18)	2.687 (17.12)
40%	2.178 (24.28)	2.196 (24.33)	2.235 (24.67)	2.216 (24.48)	2.292 (27.38)	2.782 (24.45)	2.803 (24.47)	2.854 (24.83)	2.831 (24.67)	2.927 (27.55)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.760	1.801	1.793	1.827	1.835	2.237	2.290	2.279	2.324	2.334
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.880 (6.86)	1.854 (2.94)	1.889 (5.39)	1.903 (4.12)	1.911 (4.09)	2.397 (7.15)	2.364 (3.21)	2.409 (5.67)	2.426 (4.40)	2.436 (4.36)
20%	1.959 (11.35)	1.973 (9.56)	1.985 (10.72)	2.029 (11.04)	2.021 (10.13)	2.499 (11.68)	2.517 (9.89)	2.531 (11.06)	2.588 (11.37)	2.578 (10.47)
30%	2.119 (20.40)	2.085 (15.78)	2.138 (19.25)	2.163 (18.37)	2.153 (17.30)	2.704 (20.87)	2.662 (16.23)	2.729 (19.71)	2.760 (18.80)	2.748 (17.73)
40%	2.292 (30.27)	2.311 (28.34)	2.267 (26.44)	2.318 (26.86)	2.301 (25.35)	2.932 (31.06)	2.957 (29.10)	2.901 (27.29)	2.966 (27.63)	2.945 (26.19)
Imputation using predictive mean matching method										
10%	1.847 (4.96)	1.830 (1.62)	1.877 (4.69)	1.883 (3.07)	1.888 (2.86)	2.355 (5.26)	2.334 (1.91)	2.393 (4.98)	2.402 (3.36)	2.408 (3.15)
20%	1.905 (8.26)	1.912 (6.17)	1.936 (8.00)	1.971 (7.86)	1.977 (7.73)	2.431 (8.67)	2.441 (6.58)	2.471 (8.39)	2.515 (8.26)	2.524 (8.13)
30%	2.018 (14.68)	1.979 (9.88)	2.061 (14.98)	2.078 (13.72)	2.043 (11.32)	2.574 (15.05)	2.525 (10.24)	2.628 (15.31)	2.650 (14.04)	2.606 (11.67)
40%	2.172 (23.44)	2.174 (20.71)	2.192 (22.29)	2.205 (20.69)	2.215 (20.69)	2.770 (23.80)	2.773 (21.07)	2.796 (22.67)	2.813 (21.07)	2.826 (21.10)
Imputation using propensity−score method										
10%	1.862 (5.81)	1.841 (2.22)	1.880 (4.86)	1.893 (3.57)	1.901 (3.56)	2.374 (6.10)	2.347 (2.49)	2.396 (5.14)	2.413 (3.85)	2.424 (3.83)
20%	1.936 (10.05)	1.950 (8.28)	1.954 (8.98)	1.994 (9.14)	1.999 (8.92)	2.470 (10.39)	2.487 (8.61)	2.492 (9.31)	2.544 (9.47)	2.550 (9.25)
30%	2.061 (17.12)	2.054 (14.07)	2.089 (16.55)	2.124 (16.23)	2.092 (13.96)	2.630 (17.55)	2.621 (14.44)	2.666 (16.95)	2.709 (16.60)	2.669 (14.34)
40%	2.190 (24.48)	2.213 (22.88)	2.221 (23.92)	2.239 (22.55)	2.254 (22.80)	2.802 (25.24)	2.831 (23.60)	2.842 (24.67)	2.865 (23.29)	2.884 (23.56)

SampleSize_n100_5Treatments/table3_shrink_5_treatments_MAR_n100

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.753	1.766	1.793	1.781	1.799	2.256	2.257	2.264	2.264	2.265
Imputation using linear regression										
10%	1.838 (4.87)	1.842 (4.30)	1.897 (5.80)	1.875 (5.31)	1.901 (5.63)	2.356 (4.44)	2.361 (4.62)	2.388 (5.47)	2.375 (4.91)	2.388 (5.41)
20%	1.928 (10.02)	1.944 (10.10)	1.978 (10.35)	1.981 (11.23)	1.988 (10.50)	2.482 (10.02)	2.489 (10.30)	2.523 (11.43)	2.513 (10.99)	2.514 (10.98)
30%	2.074 (18.33)	2.120 (20.04)	2.147 (19.77)	2.112 (18.63)	2.206 (22.63)	2.662 (18.02)	2.694 (19.38)	2.729 (20.54)	2.689 (18.76)	2.771 (22.32)
40%	2.281 (30.18)	2.262 (28.10)	2.327 (29.83)	2.312 (29.87)	2.372 (31.86)	2.922 (29.51)	2.898 (28.44)	2.964 (30.91)	2.942 (29.93)	3.003 (32.57)
Imputation using predictive mean matching method										
10%	1.824 (4.10)	1.835 (3.92)	1.882 (4.96)	1.863 (4.62)	1.897 (5.44)	2.335 (3.51)	2.349 (4.11)	2.362 (4.32)	2.355 (4.03)	2.378 (4.96)
20%	1.875 (7.01)	1.898 (7.48)	1.929 (7.60)	1.920 (7.80)	1.966 (9.26)	2.422 (7.34)	2.433 (7.83)	2.461 (8.71)	2.437 (7.64)	2.483 (9.59)
30%	1.958 (11.71)	2.039 (15.44)	2.073 (15.65)	2.017 (13.27)	2.088 (16.07)	2.518 (11.63)	2.587 (14.64)	2.636 (16.42)	2.565 (13.29)	2.620 (15.67)
40%	2.152 (22.77)	2.165 (22.62)	2.231 (24.43)	2.192 (23.09)	2.261 (25.66)	2.746 (21.71)	2.766 (22.56)	2.831 (25.07)	2.776 (22.62)	2.848 (25.73)
Imputation using propensity−score method										
10%	1.830 (4.42)	1.836 (3.99)	1.888 (5.32)	1.857 (4.31)	1.895 (5.32)	2.345 (3.93)	2.353 (4.28)	2.374 (4.85)	2.351 (3.82)	2.378 (4.98)
20%	1.892 (7.97)	1.920 (8.72)	1.952 (8.91)	1.934 (8.63)	1.983 (10.23)	2.439 (8.09)	2.459 (8.98)	2.490 (10.00)	2.456 (8.48)	2.505 (10.58)
30%	1.987 (13.36)	2.052 (16.19)	2.048 (14.22)	2.049 (15.09)	2.105 (17.00)	2.559 (13.43)	2.608 (15.55)	2.606 (15.09)	2.607 (15.15)	2.645 (16.77)
40%	2.178 (24.28)	2.196 (24.33)	2.235 (24.67)	2.216 (24.48)	2.292 (27.38)	2.787 (23.56)	2.812 (24.63)	2.844 (25.63)	2.817 (24.43)	2.898 (27.93)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.760	1.801	1.793	1.827	1.835	2.244	2.260	2.259	2.280	2.285
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.880 (6.86)	1.854 (2.94)	1.889 (5.39)	1.903 (4.12)	1.911 (4.09)	2.375 (5.83)	2.361 (4.45)	2.379 (5.30)	2.381 (4.43)	2.387 (4.45)
20%	1.959 (11.35)	1.973 (9.56)	1.985 (10.72)	2.029 (11.04)	2.021 (10.13)	2.502 (11.47)	2.513 (11.18)	2.520 (11.53)	2.549 (11.83)	2.542 (11.25)
30%	2.119 (20.40)	2.085 (15.78)	2.138 (19.25)	2.163 (18.37)	2.153 (17.30)	2.690 (19.87)	2.664 (17.84)	2.709 (19.87)	2.739 (20.16)	2.729 (19.44)
40%	2.292 (30.27)	2.311 (28.34)	2.267 (26.44)	2.318 (26.86)	2.301 (25.35)	2.933 (30.71)	2.954 (30.68)	2.895 (28.11)	2.959 (29.78)	2.926 (28.05)
Imputation using predictive mean matching method										
10%	1.847 (4.96)	1.830 (1.62)	1.877 (4.69)	1.883 (3.07)	1.888 (2.86)	2.335 (4.04)	2.332 (3.18)	2.359 (4.41)	2.357 (3.37)	2.356 (3.13)
20%	1.905 (8.26)	1.912 (6.17)	1.936 (8.00)	1.971 (7.86)	1.977 (7.73)	2.435 (8.48)	2.441 (7.98)	2.462 (8.96)	2.475 (8.58)	2.487 (8.85)
30%	2.018 (14.68)	1.979 (9.88)	2.061 (14.98)	2.078 (13.72)	2.043 (11.32)	2.563 (14.20)	2.525 (11.71)	2.610 (15.50)	2.630 (15.39)	2.586 (13.16)
40%	2.172 (23.44)	2.174 (20.71)	2.192 (22.29)	2.205 (20.69)	2.215 (20.69)	2.773 (23.56)	2.771 (22.61)	2.786 (23.28)	2.805 (23.04)	2.802 (22.65)
Imputation using propensity−score method										
10%	1.862 (5.81)	1.841 (2.22)	1.880 (4.86)	1.893 (3.57)	1.901 (3.56)	2.351 (4.77)	2.345 (3.75)	2.365 (4.66)	2.368 (3.86)	2.373 (3.86)
20%	1.936 (10.05)	1.950 (8.28)	1.954 (8.98)	1.994 (9.14)	1.999 (8.92)	2.474 (10.24)	2.486 (9.97)	2.484 (9.93)	2.505 (9.89)	2.514 (10.01)
30%	2.061 (17.12)	2.054 (14.07)	2.089 (16.55)	2.124 (16.23)	2.092 (13.96)	2.614 (16.48)	2.621 (15.97)	2.647 (17.15)	2.689 (17.95)	2.649 (15.93)
40%	2.190 (24.48)	2.213 (22.88)	2.221 (23.92)	2.239 (22.55)	2.254 (22.80)	2.803 (24.92)	2.826 (25.03)	2.832 (25.32)	2.856 (25.29)	2.864 (25.34)

SampleSize_n150_5Treatments/table2_nonshrink_5_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	96.55	100.00	96.55	96.55	82.76	96.55	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
30%	86.21	86.21	96.55	82.76	93.10	68.97	93.10	96.55	96.55	96.55	100.00
40%	96.43	96.43	96.43	96.43	100.00	89.29	100.00	100.00	96.43	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	92.86	92.86	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
30%	82.76	89.66	96.55	82.76	100.00	72.41	93.10	96.55	96.55	93.10	100.00
40%	100.00	96.43	96.43	89.29	92.86	75.00	100.00	96.43	96.43	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	96.43	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	96.55	100.00	100.00
30%	82.76	93.10	96.55	79.31	93.10	72.41	93.10	96.55	96.55	89.66	100.00
40%	96.43	96.43	92.86	92.86	96.43	85.71	100.00	96.43	96.43	100.00	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	92.86	100.00	92.86	92.86	75.00	100.00	96.43	100.00	92.86	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_5Treatments/table2_nonshrink_5_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	92.86	92.86	100.00	92.86	100.00	82.14	100.00	96.43	100	96.43	100.00
20%	89.66	89.66	96.55	93.10	100.00	72.41	100.00	96.55	100	96.55	100.00
30%	100.00	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100	96.55	100.00
40%	96.43	92.86	100.00	96.43	89.29	85.71	100.00	100.00	100	100.00	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	92.86	100.00	82.14	100.00	96.43	100	96.43	100.00
20%	93.10	89.66	96.55	93.10	100.00	79.31	100.00	100.00	100	93.10	100.00
30%	100.00	100.00	93.10	96.55	100.00	89.66	100.00	100.00	100	96.55	100.00
40%	92.86	89.29	96.43	96.43	89.29	78.57	100.00	100.00	100	96.43	92.86
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	96.43	92.86	100.00	78.57	96.43	96.43	100	96.43	100.00
20%	93.10	86.21	96.55	93.10	100.00	75.86	100.00	100.00	100	93.10	100.00
30%	100.00	100.00	96.55	96.55	100.00	93.10	100.00	100.00	100	96.55	100.00
40%	96.43	96.43	100.00	92.86	89.29	82.14	100.00	100.00	100	100.00	96.43

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	96.55
Imputation using linear regression	NA
10%	92.86
20%	100.00
30%	89.66
40%	96.43
Imputation using predictive mean matching method	NA
10%	92.86
20%	100.00
30%	89.66
40%	92.86
Imputation using propensity−score method	NA
10%	92.86
20%	96.55
30%	86.21
40%	92.86
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	92.86
Imputation using linear regression	NA

MR	Joint_many
10%	92.86
20%	93.10
30%	96.55
40%	96.43
Imputation using predictive mean matching method	NA
10%	92.86
20%	93.10
30%	96.55
40%	89.29
Imputation using propensity−score method	NA
10%	92.86
20%	93.10
30%	96.55
40%	96.43

SampleSize_n150_5Treatments/table2_shrink_5_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	96.55	100.00	96.55	96.55	82.76	96.55	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
30%	86.21	86.21	96.55	82.76	93.10	68.97	93.10	96.55	96.55	96.55	100.00
40%	96.43	96.43	96.43	96.43	100.00	89.29	100.00	100.00	96.43	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	92.86	92.86	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
30%	82.76	89.66	96.55	82.76	100.00	72.41	93.10	96.55	96.55	93.10	100.00
40%	100.00	96.43	96.43	89.29	92.86	75.00	100.00	96.43	96.43	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	96.43	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	96.55	100.00	100.00
30%	82.76	93.10	96.55	79.31	93.10	72.41	93.10	96.55	96.55	89.66	100.00
40%	96.43	96.43	92.86	92.86	96.43	85.71	100.00	96.43	96.43	100.00	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	92.86	100.00	92.86	92.86	75.00	100.00	96.43	100.00	92.86	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_5Treatments/table2_shrink_5_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	92.86	92.86	100.00	92.86	100.00	82.14	100.00	96.43	100	96.43	100.00
20%	89.66	89.66	96.55	93.10	100.00	72.41	100.00	96.55	100	93.10	100.00
30%	100.00	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100	96.55	100.00
40%	96.43	92.86	100.00	96.43	89.29	85.71	100.00	100.00	100	100.00	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	92.86	100.00	82.14	100.00	96.43	100	96.43	100.00
20%	93.10	89.66	96.55	93.10	100.00	79.31	100.00	96.55	100	93.10	100.00
30%	100.00	100.00	93.10	96.55	100.00	89.66	100.00	100.00	100	96.55	100.00
40%	92.86	89.29	96.43	96.43	89.29	78.57	100.00	100.00	100	96.43	92.86
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	96.43	92.86	100.00	78.57	96.43	96.43	100	96.43	100.00
20%	93.10	86.21	96.55	93.10	100.00	75.86	100.00	100.00	100	93.10	100.00
30%	100.00	100.00	96.55	96.55	100.00	93.10	100.00	100.00	100	96.55	100.00
40%	96.43	96.43	100.00	92.86	89.29	82.14	100.00	100.00	100	100.00	96.43

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	96.55
Imputation using linear regression	NA
10%	92.86
20%	100.00
30%	89.66
40%	96.43
Imputation using predictive mean matching method	NA
10%	92.86
20%	100.00
30%	89.66
40%	92.86
Imputation using propensity−score method	NA
10%	92.86
20%	96.55
30%	86.21
40%	92.86
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	92.86
Imputation using linear regression	NA

MR	Joint_many
10%	92.86
20%	89.66
30%	96.55
40%	96.43
Imputation using predictive mean matching method	NA
10%	92.86
20%	89.66
30%	96.55
40%	89.29
Imputation using propensity−score method	NA
10%	92.86
20%	93.10
30%	96.55
40%	96.43

SampleSize_n150_5Treatments/table3_nonshrink_5_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.434	1.448	1.460	1.480	1.500	1.827	1.845	1.860	1.885	1.912
Imputation using linear regression										
10%	1.508 (5.18)	1.519 (4.94)	1.520 (4.15)	1.527 (3.18)	1.558 (3.88)	1.925 (5.38)	1.940 (5.15)	1.941 (4.35)	1.949 (3.39)	1.990 (4.09)
20%	1.577 (9.96)	1.600 (10.51)	1.589 (8.82)	1.620 (9.51)	1.644 (9.59)	2.015 (10.30)	2.045 (10.84)	2.031 (9.17)	2.071 (9.86)	2.101 (9.93)
30%	1.695 (18.22)	1.725 (19.14)	1.739 (19.12)	1.759 (18.90)	1.808 (20.53)	2.165 (18.48)	2.202 (19.38)	2.221 (19.40)	2.247 (19.16)	2.308 (20.76)
40%	1.820 (26.94)	1.851 (27.88)	1.851 (26.82)	1.871 (26.43)	1.918 (27.85)	2.328 (27.40)	2.367 (28.30)	2.368 (27.29)	2.393 (26.89)	2.452 (28.29)
Imputation using predictive mean matching method										
10%	1.499 (4.53)	1.506 (4.05)	1.511 (3.52)	1.520 (2.72)	1.546 (3.08)	1.913 (4.70)	1.923 (4.23)	1.929 (3.69)	1.940 (2.90)	1.974 (3.26)
20%	1.549 (8.03)	1.575 (8.77)	1.573 (7.75)	1.612 (8.95)	1.619 (7.92)	1.979 (8.32)	2.012 (9.06)	2.010 (8.04)	2.059 (9.23)	2.068 (8.20)
30%	1.623 (13.18)	1.633 (12.82)	1.672 (14.55)	1.688 (14.04)	1.728 (15.16)	2.074 (13.49)	2.087 (13.15)	2.137 (14.88)	2.156 (14.36)	2.207 (15.47)
40%	1.754 (22.32)	1.744 (20.47)	1.772 (21.41)	1.813 (22.52)	1.824 (21.62)	2.241 (22.64)	2.228 (20.79)	2.265 (21.76)	2.316 (22.85)	2.331 (21.94)
Imputation using propensity−score method										
10%	1.501 (4.70)	1.508 (4.18)	1.512 (3.58)	1.523 (2.91)	1.553 (3.53)	1.917 (4.90)	1.925 (4.38)	1.930 (3.78)	1.944 (3.11)	1.983 (3.73)
20%	1.571 (9.59)	1.598 (10.40)	1.592 (9.06)	1.614 (9.10)	1.650 (9.96)	2.008 (9.88)	2.042 (10.68)	2.034 (9.34)	2.062 (9.39)	2.107 (10.24)
30%	1.659 (15.73)	1.647 (13.74)	1.710 (17.15)	1.724 (16.52)	1.741 (16.09)	2.121 (16.09)	2.106 (14.15)	2.186 (17.52)	2.204 (16.88)	2.226 (16.46)
40%	1.794 (25.12)	1.818 (25.61)	1.819 (24.63)	1.843 (24.54)	1.883 (25.55)	2.296 (25.68)	2.327 (26.12)	2.329 (25.21)	2.359 (25.12)	2.410 (26.10)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.433	1.450	1.455	1.471	1.490	1.824	1.846	1.853	1.874	1.897
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.477 (3.07)	1.503 (3.68)	1.526 (4.82)	1.546 (5.09)	1.554 (4.35)	1.885 (3.34)	1.919 (3.95)	1.948 (5.10)	1.975 (5.37)	1.985 (4.63)
20%	1.586 (10.69)	1.602 (10.49)	1.625 (11.62)	1.646 (11.83)	1.629 (9.35)	2.024 (10.93)	2.044 (10.71)	2.073 (11.87)	2.100 (12.08)	2.079 (9.59)
30%	1.683 (17.47)	1.728 (19.22)	1.723 (18.37)	1.714 (16.47)	1.750 (17.49)	2.147 (17.68)	2.205 (19.43)	2.198 (18.60)	2.187 (16.72)	2.233 (17.72)
40%	1.828 (27.57)	1.823 (25.77)	1.866 (28.21)	1.851 (25.78)	1.923 (29.08)	2.332 (27.84)	2.328 (26.08)	2.381 (28.44)	2.362 (26.07)	2.453 (29.33)
Imputation using predictive mean matching method										
10%	1.466 (2.33)	1.488 (2.62)	1.513 (3.93)	1.535 (4.33)	1.534 (2.95)	1.872 (2.62)	1.900 (2.91)	1.932 (4.24)	1.961 (4.63)	1.959 (3.25)
20%	1.555 (8.58)	1.571 (8.39)	1.590 (9.22)	1.612 (9.56)	1.608 (7.99)	1.987 (8.93)	2.007 (8.72)	2.031 (9.56)	2.060 (9.91)	2.055 (8.32)
30%	1.611 (12.42)	1.672 (15.31)	1.664 (14.33)	1.668 (13.35)	1.685 (13.16)	2.056 (12.71)	2.134 (15.59)	2.124 (14.61)	2.129 (13.62)	2.152 (13.44)
40%	1.720 (20.03)	1.740 (20.04)	1.765 (21.28)	1.790 (21.64)	1.818 (22.07)	2.195 (20.32)	2.221 (20.29)	2.252 (21.50)	2.284 (21.87)	2.321 (22.35)
Imputation using propensity−score method										
10%	1.472 (2.78)	1.496 (3.17)	1.520 (4.41)	1.543 (4.84)	1.546 (3.78)	1.880 (3.08)	1.910 (3.46)	1.941 (4.70)	1.970 (5.14)	1.974 (4.07)
20%	1.576 (10.02)	1.575 (8.66)	1.594 (9.53)	1.633 (11.00)	1.629 (9.37)	2.013 (10.32)	2.011 (8.94)	2.036 (9.82)	2.085 (11.29)	2.080 (9.64)
30%	1.645 (14.83)	1.685 (16.21)	1.698 (16.65)	1.693 (15.07)	1.711 (14.87)	2.099 (15.06)	2.150 (16.44)	2.166 (16.87)	2.160 (15.30)	2.184 (15.11)
40%	1.822 (27.17)	1.798 (24.00)	1.829 (25.68)	1.844 (25.33)	1.884 (26.45)	2.323 (27.35)	2.294 (24.24)	2.333 (25.89)	2.353 (25.56)	2.403 (26.68)

SampleSize_n150_5Treatments/table3_shrink_5_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.434	1.448	1.460	1.480	1.500	1.837	1.845	1.852	1.861	1.869
Imputation using linear regression										
10%	1.508 (5.18)	1.519 (4.94)	1.520 (4.15)	1.527 (3.18)	1.558 (3.88)	1.922 (4.62)	1.936 (4.94)	1.928 (4.10)	1.927 (3.54)	1.947 (4.19)
20%	1.577 (9.96)	1.600 (10.51)	1.589 (8.82)	1.620 (9.51)	1.644 (9.59)	2.026 (10.26)	2.039 (10.54)	2.029 (9.58)	2.056 (10.51)	2.073 (10.92)
30%	1.695 (18.22)	1.725 (19.14)	1.739 (19.12)	1.759 (18.90)	1.808 (20.53)	2.170 (18.13)	2.202 (19.38)	2.207 (19.18)	2.232 (19.93)	2.284 (22.23)
40%	1.820 (26.94)	1.851 (27.88)	1.851 (26.82)	1.871 (26.43)	1.918 (27.85)	2.340 (27.37)	2.375 (28.77)	2.365 (27.68)	2.385 (28.17)	2.439 (30.52)
Imputation using predictive mean matching method										
10%	1.499 (4.53)	1.506 (4.05)	1.511 (3.52)	1.520 (2.72)	1.546 (3.08)	1.911 (4.00)	1.918 (3.96)	1.916 (3.45)	1.917 (3.03)	1.930 (3.29)
20%	1.549 (8.03)	1.575 (8.77)	1.573 (7.75)	1.612 (8.95)	1.619 (7.92)	1.990 (8.33)	2.006 (8.76)	2.008 (8.44)	2.040 (9.64)	2.038 (9.05)
30%	1.623 (13.18)	1.633 (12.82)	1.672 (14.55)	1.688 (14.04)	1.728 (15.16)	2.081 (13.30)	2.089 (13.25)	2.124 (14.68)	2.144 (15.19)	2.183 (16.84)
40%	1.754 (22.32)	1.744 (20.47)	1.772 (21.41)	1.813 (22.52)	1.824 (21.62)	2.251 (22.56)	2.239 (21.37)	2.262 (22.13)	2.309 (24.06)	2.319 (24.12)
Imputation using propensity−score method										
10%	1.501 (4.70)	1.508 (4.18)	1.512 (3.58)	1.523 (2.91)	1.553 (3.53)	1.915 (4.25)	1.921 (4.15)	1.918 (3.56)	1.923 (3.33)	1.941 (3.87)
20%	1.571 (9.59)	1.598 (10.40)	1.592 (9.06)	1.614 (9.10)	1.650 (9.96)	2.018 (9.84)	2.038 (10.51)	2.033 (9.76)	2.047 (9.98)	2.079 (11.25)
30%	1.659 (15.73)	1.647 (13.74)	1.710 (17.15)	1.724 (16.52)	1.741 (16.09)	2.127 (15.77)	2.108 (14.26)	2.174 (17.37)	2.190 (17.67)	2.203 (17.90)
40%	1.794 (25.12)	1.818 (25.61)	1.819 (24.63)	1.843 (24.54)	1.883 (25.55)	2.306 (25.54)	2.335 (26.59)	2.326 (25.58)	2.353 (26.43)	2.400 (28.45)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.433	1.450	1.455	1.471	1.490	1.828	1.836	1.841	1.848	1.858
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.477 (3.07)	1.503 (3.68)	1.526 (4.82)	1.546 (5.09)	1.554 (4.35)	1.900 (3.90)	1.919 (4.53)	1.931 (4.90)	1.943 (5.19)	1.954 (5.14)
20%	1.586 (10.69)	1.602 (10.49)	1.625 (11.62)	1.646 (11.83)	1.629 (9.35)	2.028 (10.93)	2.044 (11.31)	2.052 (11.46)	2.070 (12.05)	2.053 (10.49)
30%	1.683 (17.47)	1.728 (19.22)	1.723 (18.37)	1.714 (16.47)	1.750 (17.49)	2.152 (17.72)	2.190 (19.30)	2.183 (18.60)	2.171 (17.50)	2.209 (18.89)
40%	1.828 (27.57)	1.823 (25.77)	1.866 (28.21)	1.851 (25.78)	1.923 (29.08)	2.336 (27.75)	2.326 (26.69)	2.378 (29.16)	2.356 (27.51)	2.435 (31.04)
Imputation using predictive mean matching method										
10%	1.466 (2.33)	1.488 (2.62)	1.513 (3.93)	1.535 (4.33)	1.534 (2.95)	1.887 (3.22)	1.901 (3.52)	1.914 (4.00)	1.929 (4.41)	1.928 (3.74)
20%	1.555 (8.58)	1.571 (8.39)	1.590 (9.22)	1.612 (9.56)	1.608 (7.99)	1.992 (8.95)	2.006 (9.27)	2.011 (9.25)	2.028 (9.78)	2.029 (9.15)
30%	1.611 (12.42)	1.672 (15.31)	1.664 (14.33)	1.668 (13.35)	1.685 (13.16)	2.062 (12.80)	2.117 (15.28)	2.107 (14.44)	2.114 (14.41)	2.125 (14.35)
40%	1.720 (20.03)	1.740 (20.04)	1.765 (21.28)	1.790 (21.64)	1.818 (22.07)	2.199 (20.27)	2.219 (20.87)	2.251 (22.27)	2.276 (23.21)	2.302 (23.85)
Imputation using propensity−score method										
10%	1.472 (2.78)	1.496 (3.17)	1.520 (4.41)	1.543 (4.84)	1.546 (3.78)	1.895 (3.64)	1.911 (4.08)	1.924 (4.53)	1.939 (4.93)	1.944 (4.60)
20%	1.576 (10.02)	1.575 (8.66)	1.594 (9.53)	1.633 (11.00)	1.629 (9.37)	2.015 (10.21)	2.013 (9.61)	2.016 (9.52)	2.056 (11.27)	2.057 (10.67)
30%	1.645 (14.83)	1.685 (16.21)	1.698 (16.65)	1.693 (15.07)	1.711 (14.87)	2.105 (15.15)	2.134 (16.24)	2.151 (16.84)	2.144 (16.06)	2.158 (16.13)
40%	1.822 (27.17)	1.798 (24.00)	1.829 (25.68)	1.844 (25.33)	1.884 (26.45)	2.327 (27.28)	2.293 (24.87)	2.331 (26.64)	2.347 (27.01)	2.386 (28.41)

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	93.10	100.00	100.00	93.10	100.00	100.00	96.55	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	79.31	58.62	93.10	96.55	96.55	96.55	96.55
20%	92.86	100.00	100.00	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
30%	100.00	100.00	96.43	96.43	100.00	92.86	100.00	100.00	100.00	100.00	100.00
40%	100.00	100.00	100.00	93.10	100.00	93.10	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	82.76	62.07	93.10	96.55	96.55	96.55	96.55
20%	85.71	100.00	100.00	96.43	92.86	75.00	96.43	100.00	100.00	100.00	92.86
30%	89.29	92.86	96.43	96.43	100.00	75.00	100.00	100.00	100.00	100.00	100.00
40%	96.55	100.00	96.55	93.10	96.55	82.76	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	79.31	58.62	93.10	96.55	96.55	96.55	93.10
20%	89.29	92.86	100.00	96.43	89.29	71.43	92.86	100.00	100.00	100.00	96.43
30%	89.29	92.86	96.43	92.86	96.43	71.43	96.43	100.00	100.00	100.00	100.00
40%	89.66	93.10	93.10	86.21	93.10	75.86	100.00	96.55	96.55	89.66	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	96.55	100.00	96.55	96.55	89.66	100.00	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_5Treatments/table2_nonshrink_5_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.55	86.21	89.66	96.55	96.55	75.86	96.55	96.55	96.55	96.55	100.00
20%	92.86	92.86	96.43	92.86	85.71	75.00	96.43	100.00	96.43	92.86	96.43
30%	93.10	96.55	96.55	100.00	100.00	89.66	96.55	100.00	100.00	100.00	100.00
40%	96.55	89.66	96.55	100.00	93.10	79.31	100.00	96.55	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	93.10	93.10	68.97	96.55	96.55	96.55	96.55	100.00
20%	92.86	89.29	96.43	89.29	85.71	71.43	92.86	100.00	96.43	92.86	96.43
30%	89.66	93.10	93.10	96.55	96.55	75.86	96.55	96.55	96.55	100.00	100.00
40%	93.10	82.76	89.66	93.10	89.66	68.97	96.55	93.10	96.55	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	96.55	93.10	72.41	96.55	96.55	96.55	96.55	100.00
20%	92.86	89.29	96.43	89.29	85.71	71.43	92.86	100.00	96.43	92.86	96.43
30%	89.66	96.55	93.10	93.10	100.00	75.86	96.55	100.00	100.00	100.00	100.00
40%	93.10	82.76	89.66	89.66	82.76	62.07	96.55	89.66	100.00	100.00	93.10

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	96.55
Imputation using linear regression	NA
10%	93.10
20%	92.86
30%	100.00
40%	100.00
Imputation using predictive mean matching method	NA
10%	93.10
20%	89.29
30%	100.00
40%	96.55
Imputation using propensity−score method	NA
10%	89.66
20%	89.29
30%	96.43
40%	82.76
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	89.29
30%	96.55
40%	96.55
Imputation using predictive mean matching method	NA
10%	89.66
20%	85.71
30%	89.66
40%	89.66
Imputation using propensity−score method	NA
10%	89.66
20%	85.71
30%	96.55
40%	86.21

SampleSize_n20_5Treatments/table2_shrink_5_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	93.10	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	79.31	58.62	96.55	96.55	100.00	96.55	96.55
20%	92.86	100.00	100.00	96.43	96.43	85.71	100.00	100.00	100.00	100.00	100.00
30%	100.00	100.00	96.43	96.43	100.00	92.86	100.00	100.00	100.00	100.00	100.00
40%	100.00	100.00	100.00	93.10	100.00	93.10	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	82.76	62.07	96.55	96.55	100.00	96.55	93.10
20%	85.71	100.00	100.00	96.43	92.86	75.00	100.00	100.00	100.00	100.00	96.43
30%	89.29	92.86	96.43	96.43	100.00	75.00	100.00	100.00	100.00	100.00	100.00
40%	96.55	100.00	96.55	93.10	96.55	82.76	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	79.31	58.62	93.10	96.55	96.55	96.55	93.10
20%	89.29	92.86	100.00	96.43	89.29	71.43	96.43	100.00	100.00	100.00	96.43
30%	89.29	92.86	96.43	92.86	96.43	71.43	100.00	100.00	100.00	100.00	96.43
40%	89.66	93.10	93.10	86.21	93.10	75.86	96.55	96.55	96.55	93.10	96.55
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	96.55	100.00	96.55	96.55	89.66	100.00	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_5Treatments/table2_shrink_5_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.55	86.21	89.66	96.55	96.55	75.86	96.55	96.55	96.55	96.55	100.00
20%	92.86	92.86	96.43	92.86	85.71	75.00	96.43	100.00	96.43	92.86	96.43
30%	93.10	96.55	96.55	100.00	100.00	89.66	96.55	100.00	100.00	100.00	100.00
40%	96.55	89.66	96.55	100.00	93.10	79.31	100.00	96.55	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	93.10	93.10	68.97	96.55	96.55	96.55	96.55	100.00
20%	92.86	89.29	96.43	89.29	85.71	71.43	92.86	100.00	96.43	92.86	96.43
30%	89.66	93.10	93.10	96.55	96.55	75.86	96.55	96.55	100.00	100.00	100.00
40%	93.10	82.76	89.66	93.10	89.66	68.97	96.55	93.10	96.55	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	96.55	93.10	72.41	96.55	96.55	96.55	96.55	100.00
20%	92.86	89.29	96.43	89.29	85.71	71.43	92.86	100.00	96.43	92.86	96.43
30%	89.66	96.55	93.10	93.10	100.00	75.86	96.55	100.00	100.00	100.00	100.00
40%	93.10	82.76	89.66	89.66	82.76	62.07	96.55	89.66	100.00	96.55	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	100.00
Imputation using linear regression	NA
10%	96.55
20%	100.00
30%	100.00
40%	100.00
Imputation using predictive mean matching method	NA
10%	93.10
20%	96.43
30%	100.00
40%	96.55
Imputation using propensity−score method	NA
10%	89.66
20%	92.86
30%	96.43
40%	86.21
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	89.29
30%	96.55
40%	96.55
Imputation using predictive mean matching method	NA
10%	89.66
20%	85.71
30%	93.10
40%	93.10
Imputation using propensity−score method	NA
10%	89.66
20%	85.71
30%	96.55
40%	86.21

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	4.066	4.131	4.153	4.259	4.175	5.086	5.169	5.196	5.329	5.221
Imputation using linear regression										
10%	4.177 (2.73)	4.318 (4.51)	4.427 (6.58)	4.385 (2.97)	4.492 (7.58)	5.335 (4.90)	5.516 (6.73)	5.652 (8.78)	5.600 (5.08)	5.737 (9.87)
20%	4.480 (10.18)	4.707 (13.95)	4.633 (11.54)	4.703 (10.43)	4.808 (15.16)	5.716 (12.38)	6.007 (16.22)	5.909 (13.73)	6.000 (12.60)	6.135 (17.51)
30%	4.788 (17.76)	4.993 (20.85)	4.941 (18.98)	5.110 (19.98)	5.005 (19.87)	6.119 (20.31)	6.374 (23.32)	6.314 (21.52)	6.527 (22.48)	6.396 (22.50)
40%	6.032 (48.36)	5.911 (43.07)	6.048 (45.61)	6.056 (42.19)	6.129 (46.80)	7.648 (50.36)	7.499 (45.08)	7.673 (47.67)	7.681 (44.13)	7.767 (48.75)
Imputation using predictive mean matching method										
10%	4.028 (−0.93)	4.173 (1.00)	4.217 (1.53)	4.183 (−1.77)	4.316 (3.38)	5.140 (1.05)	5.325 (3.03)	5.379 (3.52)	5.337 (0.14)	5.507 (5.48)
20%	4.017 (−1.21)	4.241 (2.66)	4.179 (0.61)	4.234 (−0.58)	4.436 (6.24)	5.126 (0.78)	5.413 (4.72)	5.331 (2.61)	5.403 (1.39)	5.660 (8.40)
30%	4.088 (0.54)	3.987 (−3.49)	4.177 (0.57)	4.251 (−0.19)	4.260 (2.02)	5.222 (2.67)	5.091 (−1.50)	5.335 (2.69)	5.427 (1.84)	5.442 (4.24)
40%	4.616 (13.53)	4.474 (8.29)	4.591 (10.54)	4.605 (8.14)	4.634 (10.98)	5.872 (15.46)	5.695 (10.19)	5.843 (12.47)	5.863 (10.01)	5.898 (12.96)
Imputation using propensity−score method										
10%	4.006 (−1.49)	4.153 (0.52)	4.204 (1.22)	4.151 (−2.52)	4.297 (2.91)	5.112 (0.51)	5.301 (2.57)	5.364 (3.24)	5.298 (−0.59)	5.484 (5.03)
20%	4.056 (−0.26)	4.219 (2.13)	4.182 (0.68)	4.264 (0.12)	4.442 (6.40)	5.171 (1.67)	5.380 (4.08)	5.330 (2.59)	5.435 (1.99)	5.664 (8.48)
30%	4.097 (0.76)	3.996 (−3.28)	4.174 (0.49)	4.192 (−1.58)	4.329 (3.68)	5.232 (2.86)	5.100 (−1.33)	5.331 (2.60)	5.352 (0.43)	5.528 (5.89)
40%	4.403 (8.29)	4.320 (4.58)	4.342 (4.53)	4.447 (4.42)	4.479 (7.28)	5.620 (10.50)	5.514 (6.67)	5.540 (6.63)	5.676 (6.51)	5.718 (9.51)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	3.940	3.942	4.079	3.994	4.006	4.938	4.938	5.112	5.006	5.020
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	4.367 (10.84)	4.435 (12.50)	4.447 (9.02)	4.438 (11.14)	4.465 (11.47)	5.544 (12.28)	5.628 (13.98)	5.644 (10.39)	5.632 (12.50)	5.670 (12.95)
20%	4.590 (16.50)	4.731 (20.03)	4.730 (15.96)	4.912 (22.99)	4.876 (21.73)	5.824 (17.94)	6.004 (21.59)	6.009 (17.53)	6.238 (24.61)	6.192 (23.36)
30%	5.398 (36.99)	5.337 (35.39)	5.299 (29.90)	5.504 (37.81)	5.713 (42.63)	6.831 (38.32)	6.759 (36.89)	6.711 (31.27)	6.963 (39.10)	7.228 (44.00)
40%	6.002 (52.32)	6.132 (55.57)	5.978 (46.55)	5.940 (48.73)	5.830 (45.53)	7.570 (53.30)	7.734 (56.63)	7.543 (47.54)	7.497 (49.76)	7.356 (46.54)
Imputation using predictive mean matching method										
10%	4.138 (5.03)	4.231 (7.33)	4.235 (3.83)	4.235 (6.04)	4.240 (5.85)	5.258 (6.48)	5.374 (8.83)	5.379 (5.23)	5.380 (7.48)	5.388 (7.33)
20%	4.158 (5.53)	4.275 (8.45)	4.268 (4.62)	4.405 (10.31)	4.482 (11.88)	5.268 (6.68)	5.418 (9.73)	5.411 (5.85)	5.584 (11.55)	5.680 (13.16)
30%	4.429 (12.39)	4.491 (13.92)	4.463 (9.42)	4.432 (10.98)	4.673 (16.67)	5.634 (14.09)	5.714 (15.71)	5.680 (11.10)	5.636 (12.59)	5.944 (18.42)
40%	4.227 (7.27)	4.465 (13.26)	4.332 (6.19)	4.425 (10.80)	4.319 (7.82)	5.392 (9.19)	5.692 (15.28)	5.526 (8.09)	5.646 (12.80)	5.511 (9.79)
Imputation using propensity−score method										
10%	4.174 (5.94)	4.269 (8.29)	4.266 (4.57)	4.255 (6.55)	4.274 (6.69)	5.300 (7.33)	5.418 (9.73)	5.414 (5.90)	5.402 (7.91)	5.426 (8.10)
20%	4.181 (6.11)	4.301 (9.12)	4.271 (4.71)	4.464 (11.77)	4.507 (12.51)	5.297 (7.26)	5.451 (10.39)	5.415 (5.92)	5.658 (13.02)	5.712 (13.79)
30%	4.454 (13.05)	4.624 (17.31)	4.554 (11.65)	4.545 (13.81)	4.710 (17.57)	5.653 (14.47)	5.871 (18.89)	5.783 (13.12)	5.767 (15.20)	5.978 (19.09)
40%	4.194 (6.45)	4.403 (11.71)	4.354 (6.72)	4.323 (8.24)	4.272 (6.63)	5.366 (8.67)	5.633 (14.08)	5.568 (8.91)	5.532 (10.51)	5.466 (8.89)

SampleSize_n20_5Treatments/table3_shrink_5_treatments_MAR_n20

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	4.066	4.131	4.153	4.259	4.175	5.139	5.160	5.171	5.183	5.163
Imputation using linear regression										
10%	4.177 (2.73)	4.318 (4.51)	4.427 (6.58)	4.385 (2.97)	4.492 (7.58)	5.419 (5.45)	5.515 (6.88)	5.611 (8.52)	5.607 (8.17)	5.607 (8.58)
20%	4.480 (10.18)	4.707 (13.95)	4.633 (11.54)	4.703 (10.43)	4.808 (15.16)	5.805 (12.96)	5.959 (15.49)	5.903 (14.16)	5.956 (14.90)	5.944 (15.12)
30%	4.788 (17.76)	4.993 (20.85)	4.941 (18.98)	5.110 (19.98)	5.005 (19.87)	6.191 (20.48)	6.464 (25.28)	6.313 (22.09)	6.478 (24.97)	6.379 (23.54)
40%	6.032 (48.36)	5.911 (43.07)	6.048 (45.61)	6.056 (42.19)	6.129 (46.80)	7.674 (49.34)	7.526 (45.86)	7.701 (48.93)	7.674 (48.04)	7.728 (49.67)
Imputation using predictive mean matching method										
10%	4.028 (−0.93)	4.173 (1.00)	4.217 (1.53)	4.183 (−1.77)	4.316 (3.38)	5.215 (1.49)	5.316 (3.03)	5.348 (3.43)	5.350 (3.21)	5.370 (4.00)
20%	4.017 (−1.21)	4.241 (2.66)	4.179 (0.61)	4.234 (−0.58)	4.436 (6.24)	5.243 (2.04)	5.359 (3.87)	5.325 (2.99)	5.342 (3.05)	5.436 (5.28)
30%	4.088 (0.54)	3.987 (−3.49)	4.177 (0.57)	4.251 (−0.19)	4.260 (2.02)	5.276 (2.68)	5.226 (1.28)	5.338 (3.23)	5.384 (3.86)	5.389 (4.37)
40%	4.616 (13.53)	4.474 (8.29)	4.591 (10.54)	4.605 (8.14)	4.634 (10.98)	5.892 (14.65)	5.727 (10.98)	5.846 (13.05)	5.847 (12.79)	5.850 (13.29)
Imputation using propensity−score method										
10%	4.006 (−1.49)	4.153 (0.52)	4.204 (1.22)	4.151 (−2.52)	4.297 (2.91)	5.195 (1.09)	5.287 (2.47)	5.331 (3.09)	5.313 (2.51)	5.342 (3.47)
20%	4.056 (−0.26)	4.219 (2.13)	4.182 (0.68)	4.264 (0.12)	4.442 (6.40)	5.275 (2.66)	5.326 (3.22)	5.328 (3.03)	5.380 (3.80)	5.432 (5.19)
30%	4.097 (0.76)	3.996 (−3.28)	4.174 (0.49)	4.192 (−1.58)	4.329 (3.68)	5.278 (2.70)	5.246 (1.67)	5.328 (3.04)	5.317 (2.58)	5.477 (6.07)
40%	4.403 (8.29)	4.320 (4.58)	4.342 (4.53)	4.447 (4.42)	4.479 (7.28)	5.639 (9.73)	5.577 (8.07)	5.571 (7.75)	5.689 (9.74)	5.687 (10.14)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	3.940	3.942	4.079	3.994	4.006	4.983	4.989	5.026	5.008	4.987
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	4.367 (10.84)	4.435 (12.50)	4.447 (9.02)	4.438 (11.14)	4.465 (11.47)	5.536 (11.08)	5.578 (11.82)	5.569 (10.82)	5.567 (11.17)	5.560 (11.50)
20%	4.590 (16.50)	4.731 (20.03)	4.730 (15.96)	4.912 (22.99)	4.876 (21.73)	5.890 (18.19)	5.979 (19.85)	5.977 (18.93)	6.134 (22.50)	6.046 (21.24)
30%	5.398 (36.99)	5.337 (35.39)	5.299 (29.90)	5.504 (37.81)	5.713 (42.63)	6.862 (37.69)	6.755 (35.41)	6.674 (32.80)	6.959 (38.98)	7.110 (42.59)
40%	6.002 (52.32)	6.132 (55.57)	5.978 (46.55)	5.940 (48.73)	5.830 (45.53)	7.579 (52.09)	7.706 (54.46)	7.537 (49.98)	7.466 (49.11)	7.382 (48.03)
Imputation using predictive mean matching method										
10%	4.138 (5.03)	4.231 (7.33)	4.235 (3.83)	4.235 (6.04)	4.240 (5.85)	5.252 (5.39)	5.314 (6.53)	5.311 (5.68)	5.290 (5.64)	5.287 (6.02)
20%	4.158 (5.53)	4.275 (8.45)	4.268 (4.62)	4.405 (10.31)	4.482 (11.88)	5.321 (6.78)	5.377 (7.79)	5.359 (6.64)	5.463 (9.09)	5.493 (10.16)
30%	4.429 (12.39)	4.491 (13.92)	4.463 (9.42)	4.432 (10.98)	4.673 (16.67)	5.679 (13.95)	5.705 (14.36)	5.636 (12.14)	5.664 (13.12)	5.849 (17.29)
40%	4.227 (7.27)	4.465 (13.26)	4.332 (6.19)	4.425 (10.80)	4.319 (7.82)	5.480 (9.97)	5.677 (13.80)	5.559 (10.61)	5.599 (11.81)	5.528 (10.85)
Imputation using propensity−score method										
10%	4.174 (5.94)	4.269 (8.29)	4.266 (4.57)	4.255 (6.55)	4.274 (6.69)	5.288 (6.11)	5.351 (7.27)	5.352 (6.50)	5.315 (6.13)	5.327 (6.83)
20%	4.181 (6.11)	4.301 (9.12)	4.271 (4.71)	4.464 (11.77)	4.507 (12.51)	5.358 (7.52)	5.423 (8.71)	5.374 (6.93)	5.544 (10.71)	5.531 (10.92)
30%	4.454 (13.05)	4.624 (17.31)	4.554 (11.65)	4.545 (13.81)	4.710 (17.57)	5.712 (14.61)	5.857 (17.40)	5.740 (14.21)	5.757 (14.96)	5.883 (17.98)
40%	4.194 (6.45)	4.403 (11.71)	4.354 (6.72)	4.323 (8.24)	4.272 (6.63)	5.468 (9.72)	5.646 (13.18)	5.619 (11.82)	5.537 (10.57)	5.515 (10.60)

SampleSize_n200_5Treatments/table2_nonshrink_5_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	85.71	100.00	92.86	96.43	75.00	96.43	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	93.10	100.00	89.66	100.00	96.55	96.55	100.00	100.00
30%	93.10	96.55	96.55	100.00	86.21	75.86	100.00	100.00	100.00	100.00	93.10
40%	92.86	92.86	96.43	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	92.86	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	93.10	100.00	89.66	100.00	96.55	96.55	100.00	100.00
30%	89.66	96.55	100.00	100.00	79.31	68.97	96.55	96.55	100.00	100.00	93.10
40%	92.86	92.86	96.43	92.86	100.00	85.71	100.00	100.00	96.43	96.43	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	100.00
20%	100.00	96.55	96.55	89.66	100.00	86.21	100.00	96.55	96.55	100.00	100.00
30%	93.10	96.55	100.00	96.55	86.21	79.31	96.55	96.55	100.00	100.00	93.10
40%	96.43	96.43	96.43	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	85.71	85.71	92.86	92.86	89.29	64.29	92.86	92.86	100.00	100.00	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_5Treatments/table2_nonshrink_5_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.55	100.00	93.10	100.00	93.10	86.21	100.00	100.00	93.10	100	96.55
20%	96.55	96.55	89.66	100.00	86.21	79.31	100.00	96.55	96.55	100	100.00
30%	96.43	96.43	100.00	96.43	92.86	89.29	100.00	100.00	100.00	100	96.43
40%	92.86	96.43	96.43	100.00	100.00	92.86	92.86	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	100.00	93.10	96.55	93.10	82.76	100.00	100.00	93.10	100	96.55
20%	89.66	96.55	89.66	100.00	82.76	68.97	100.00	96.55	93.10	100	96.55
30%	96.43	100.00	100.00	100.00	89.29	85.71	100.00	100.00	100.00	100	100.00
40%	92.86	100.00	92.86	100.00	92.86	85.71	100.00	100.00	100.00	100	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	89.66	96.55	93.10	82.76	100.00	100.00	93.10	100	96.55
20%	96.55	96.55	89.66	100.00	82.76	75.86	100.00	100.00	93.10	100	96.55
30%	96.43	96.43	100.00	96.43	92.86	89.29	100.00	100.00	100.00	100	100.00
40%	92.86	92.86	96.43	100.00	96.43	85.71	96.43	100.00	96.43	100	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	96.43
Imputation using linear regression	NA
10%	96.43
20%	93.10
30%	93.10
40%	96.43
Imputation using predictive mean matching method	NA
10%	96.43
20%	93.10
30%	89.66
40%	92.86
Imputation using propensity−score method	NA
10%	100.00
20%	93.10
30%	89.66
40%	96.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	82.14
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	96.55
30%	96.43
40%	92.86
Imputation using predictive mean matching method	NA
10%	89.66
20%	89.66
30%	100.00
40%	100.00
Imputation using propensity−score method	NA
10%	89.66
20%	89.66
30%	100.00
40%	96.43

SampleSize_n200_5Treatments/table2_shrink_5_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	85.71	100.00	92.86	96.43	75.00	100.00	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	93.10	100.00	89.66	100.00	96.55	96.55	100.00	100.00
30%	93.10	96.55	96.55	100.00	86.21	75.86	100.00	100.00	100.00	100.00	93.10
40%	92.86	92.86	96.43	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	92.86	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	93.10	100.00	89.66	100.00	96.55	96.55	100.00	100.00
30%	89.66	96.55	100.00	100.00	79.31	68.97	96.55	96.55	100.00	100.00	93.10
40%	92.86	92.86	96.43	92.86	100.00	85.71	100.00	100.00	96.43	96.43	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	100.00
20%	100.00	96.55	96.55	89.66	100.00	86.21	100.00	96.55	96.55	100.00	100.00
30%	93.10	96.55	100.00	96.55	86.21	79.31	96.55	96.55	100.00	100.00	93.10
40%	96.43	96.43	96.43	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	85.71	85.71	92.86	92.86	89.29	64.29	96.43	92.86	100.00	100.00	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_5Treatments/table2_shrink_5_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.55	100.00	93.10	100.00	93.10	86.21	100.00	100.00	93.10	100	96.55
20%	96.55	96.55	89.66	100.00	86.21	79.31	100.00	96.55	96.55	100	96.55
30%	96.43	96.43	100.00	96.43	92.86	89.29	100.00	100.00	100.00	100	100.00
40%	92.86	96.43	96.43	100.00	100.00	92.86	92.86	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	100.00	93.10	96.55	93.10	82.76	100.00	100.00	93.10	100	96.55
20%	89.66	96.55	89.66	100.00	82.76	68.97	100.00	96.55	93.10	100	96.55
30%	96.43	100.00	100.00	100.00	89.29	85.71	100.00	100.00	100.00	100	100.00
40%	92.86	100.00	92.86	100.00	92.86	85.71	100.00	100.00	100.00	100	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	89.66	96.55	93.10	82.76	100.00	100.00	93.10	100	96.55
20%	96.55	96.55	89.66	100.00	82.76	75.86	100.00	100.00	93.10	100	96.55
30%	96.43	96.43	100.00	96.43	92.86	89.29	100.00	100.00	100.00	100	100.00
40%	92.86	92.86	96.43	100.00	96.43	85.71	96.43	100.00	96.43	100	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	100.00
Imputation using linear regression	NA
10%	96.43
20%	93.10
30%	93.10
40%	96.43
Imputation using predictive mean matching method	NA
10%	96.43
20%	93.10
30%	89.66
40%	92.86
Imputation using propensity−score method	NA
10%	100.00
20%	93.10
30%	89.66
40%	96.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	85.71
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	93.10
30%	100.00
40%	92.86
Imputation using predictive mean matching method	NA
10%	89.66
20%	89.66
30%	100.00
40%	100.00
Imputation using propensity−score method	NA
10%	89.66
20%	89.66
30%	100.00
40%	96.43

SampleSize_n200_5Treatments/table3_nonshrink_5_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.233	1.249	1.257	1.271	1.282	1.573	1.593	1.604	1.622	1.636
Imputation using linear regression										
10%	1.305 (5.83)	1.314 (5.26)	1.327 (5.57)	1.340 (5.45)	1.352 (5.48)	1.664 (5.81)	1.677 (5.24)	1.693 (5.56)	1.710 (5.43)	1.725 (5.47)
20%	1.363 (10.56)	1.401 (12.17)	1.390 (10.58)	1.423 (11.92)	1.408 (9.81)	1.741 (10.72)	1.789 (12.31)	1.776 (10.73)	1.817 (12.05)	1.799 (9.97)
30%	1.455 (17.99)	1.477 (18.28)	1.482 (17.90)	1.515 (19.17)	1.520 (18.59)	1.861 (18.33)	1.890 (18.63)	1.896 (18.23)	1.938 (19.50)	1.945 (18.92)
40%	1.567 (27.11)	1.611 (28.98)	1.588 (26.37)	1.639 (28.95)	1.651 (28.81)	2.004 (27.41)	2.060 (29.29)	2.032 (26.69)	2.096 (29.22)	2.112 (29.14)
Imputation using predictive mean matching method										
10%	1.296 (5.13)	1.308 (4.77)	1.319 (4.98)	1.336 (5.07)	1.346 (4.95)	1.653 (5.11)	1.669 (4.75)	1.683 (4.96)	1.704 (5.05)	1.716 (4.94)
20%	1.342 (8.83)	1.367 (9.47)	1.368 (8.84)	1.404 (10.45)	1.396 (8.90)	1.713 (8.89)	1.745 (9.54)	1.746 (8.89)	1.792 (10.50)	1.782 (8.95)
30%	1.413 (14.60)	1.431 (14.61)	1.438 (14.38)	1.454 (14.39)	1.459 (13.83)	1.808 (14.92)	1.831 (14.95)	1.839 (14.70)	1.861 (14.73)	1.868 (14.19)
40%	1.518 (23.11)	1.571 (25.84)	1.526 (21.38)	1.546 (21.64)	1.579 (23.19)	1.939 (23.31)	2.008 (26.06)	1.951 (21.65)	1.977 (21.89)	2.020 (23.48)
Imputation using propensity−score method										
10%	1.295 (5.04)	1.309 (4.80)	1.322 (5.20)	1.337 (5.17)	1.344 (4.83)	1.652 (5.03)	1.669 (4.79)	1.687 (5.18)	1.705 (5.15)	1.714 (4.82)
20%	1.358 (10.13)	1.390 (11.29)	1.397 (11.15)	1.416 (11.41)	1.407 (9.71)	1.733 (10.18)	1.774 (11.35)	1.783 (11.19)	1.807 (11.46)	1.795 (9.76)
30%	1.429 (15.91)	1.442 (15.44)	1.463 (16.38)	1.480 (16.41)	1.492 (16.41)	1.829 (16.28)	1.845 (15.83)	1.872 (16.73)	1.894 (16.79)	1.910 (16.78)
40%	1.551 (25.77)	1.599 (28.02)	1.577 (25.45)	1.587 (24.81)	1.616 (26.08)	1.983 (26.06)	2.044 (28.32)	2.017 (25.77)	2.029 (25.13)	2.068 (26.43)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.227	1.242	1.257	1.266	1.286	1.566	1.585	1.604	1.616	1.641
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.303 (6.22)	1.304 (4.98)	1.329 (5.72)	1.335 (5.43)	1.332 (3.59)	1.665 (6.30)	1.665 (5.06)	1.697 (5.79)	1.705 (5.51)	1.701 (3.66)
20%	1.381 (12.51)	1.386 (11.59)	1.399 (11.34)	1.400 (10.53)	1.424 (10.75)	1.762 (12.55)	1.769 (11.63)	1.786 (11.38)	1.787 (10.58)	1.818 (10.79)
30%	1.454 (18.53)	1.480 (19.12)	1.485 (18.14)	1.510 (19.23)	1.511 (17.52)	1.856 (18.53)	1.888 (19.11)	1.895 (18.13)	1.927 (19.24)	1.928 (17.52)
40%	1.588 (29.40)	1.599 (28.72)	1.641 (30.58)	1.620 (27.97)	1.660 (29.15)	2.029 (29.56)	2.043 (28.91)	2.097 (30.74)	2.072 (28.20)	2.121 (29.29)
Imputation using predictive mean matching method										
10%	1.290 (5.16)	1.290 (3.90)	1.322 (5.16)	1.327 (4.76)	1.321 (2.76)	1.648 (5.24)	1.648 (3.98)	1.688 (5.23)	1.694 (4.84)	1.687 (2.83)
20%	1.360 (10.88)	1.355 (9.13)	1.385 (10.19)	1.381 (9.09)	1.402 (9.06)	1.736 (10.87)	1.730 (9.12)	1.767 (10.17)	1.763 (9.09)	1.789 (9.05)
30%	1.419 (15.64)	1.446 (16.42)	1.450 (15.40)	1.461 (15.36)	1.478 (15.00)	1.812 (15.73)	1.846 (16.48)	1.852 (15.46)	1.866 (15.47)	1.888 (15.08)
40%	1.508 (22.86)	1.550 (24.79)	1.580 (25.75)	1.572 (24.14)	1.576 (22.59)	1.925 (22.92)	1.979 (24.82)	2.017 (25.77)	2.007 (24.20)	2.012 (22.65)
Imputation using propensity−score method										
10%	1.289 (5.08)	1.291 (3.96)	1.317 (4.78)	1.326 (4.70)	1.324 (3.00)	1.647 (5.19)	1.650 (4.07)	1.682 (4.88)	1.694 (4.81)	1.692 (3.11)
20%	1.366 (11.33)	1.364 (9.84)	1.377 (9.54)	1.381 (9.06)	1.412 (9.81)	1.744 (11.39)	1.742 (9.89)	1.758 (9.61)	1.764 (9.13)	1.802 (9.86)
30%	1.439 (17.24)	1.458 (17.39)	1.464 (16.49)	1.483 (17.13)	1.495 (16.33)	1.837 (17.33)	1.862 (17.47)	1.870 (16.57)	1.895 (17.23)	1.910 (16.42)
40%	1.543 (25.76)	1.588 (27.82)	1.594 (26.84)	1.599 (26.25)	1.643 (27.83)	1.971 (25.85)	2.027 (27.88)	2.036 (26.95)	2.042 (26.35)	2.097 (27.84)

SampleSize_n200_5Treatments/table3_shrink_5_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.233	1.249	1.257	1.271	1.282	1.583	1.589	1.591	1.598	1.604
Imputation using linear regression										
10%	1.305 (5.83)	1.314 (5.26)	1.327 (5.57)	1.340 (5.45)	1.352 (5.48)	1.666 (5.28)	1.670 (5.13)	1.675 (5.25)	1.683 (5.33)	1.690 (5.38)
20%	1.363 (10.56)	1.401 (12.17)	1.390 (10.58)	1.423 (11.92)	1.408 (9.81)	1.746 (10.29)	1.774 (11.68)	1.767 (11.02)	1.794 (12.25)	1.776 (10.71)
30%	1.455 (17.99)	1.477 (18.28)	1.482 (17.90)	1.515 (19.17)	1.520 (18.59)	1.867 (17.98)	1.883 (18.52)	1.893 (18.94)	1.921 (20.23)	1.925 (20.02)
40%	1.567 (27.11)	1.611 (28.98)	1.588 (26.37)	1.639 (28.95)	1.651 (28.81)	2.012 (27.10)	2.052 (29.12)	2.028 (27.46)	2.089 (30.71)	2.090 (30.33)
Imputation using predictive mean matching method										
10%	1.296 (5.13)	1.308 (4.77)	1.319 (4.98)	1.336 (5.07)	1.346 (4.95)	1.656 (4.62)	1.663 (4.65)	1.666 (4.72)	1.676 (4.87)	1.682 (4.87)
20%	1.342 (8.83)	1.367 (9.47)	1.368 (8.84)	1.404 (10.45)	1.396 (8.90)	1.718 (8.52)	1.731 (8.93)	1.737 (9.19)	1.767 (10.58)	1.759 (9.68)
30%	1.413 (14.60)	1.431 (14.61)	1.438 (14.38)	1.454 (14.39)	1.459 (13.83)	1.814 (14.58)	1.827 (15.00)	1.840 (15.62)	1.845 (15.47)	1.850 (15.33)
40%	1.518 (23.11)	1.571 (25.84)	1.526 (21.38)	1.546 (21.64)	1.579 (23.19)	1.948 (23.05)	1.999 (25.81)	1.946 (22.32)	1.970 (23.31)	1.996 (24.45)
Imputation using propensity−score method										
10%	1.295 (5.04)	1.309 (4.80)	1.322 (5.20)	1.337 (5.17)	1.344 (4.83)	1.654 (4.51)	1.662 (4.62)	1.669 (4.92)	1.679 (5.10)	1.679 (4.70)
20%	1.358 (10.13)	1.390 (11.29)	1.397 (11.15)	1.416 (11.41)	1.407 (9.71)	1.737 (9.77)	1.760 (10.77)	1.773 (11.45)	1.783 (11.62)	1.772 (10.50)
30%	1.429 (15.91)	1.442 (15.44)	1.463 (16.38)	1.480 (16.41)	1.492 (16.41)	1.835 (15.95)	1.841 (15.89)	1.871 (17.56)	1.877 (17.48)	1.891 (17.93)
40%	1.551 (25.77)	1.599 (28.02)	1.577 (25.45)	1.587 (24.81)	1.616 (26.08)	1.989 (25.69)	2.036 (28.15)	2.012 (26.47)	2.022 (26.54)	2.047 (27.64)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.227	1.242	1.257	1.266	1.286	1.582	1.584	1.590	1.596	1.604
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.303 (6.22)	1.304 (4.98)	1.329 (5.72)	1.335 (5.43)	1.332 (3.59)	1.664 (5.22)	1.664 (5.07)	1.679 (5.57)	1.683 (5.42)	1.679 (4.66)
20%	1.381 (12.51)	1.386 (11.59)	1.399 (11.34)	1.400 (10.53)	1.424 (10.75)	1.759 (11.21)	1.768 (11.64)	1.772 (11.42)	1.769 (10.86)	1.790 (11.56)
30%	1.454 (18.53)	1.480 (19.12)	1.485 (18.14)	1.510 (19.23)	1.511 (17.52)	1.859 (17.53)	1.883 (18.89)	1.884 (18.46)	1.905 (19.38)	1.906 (18.83)
40%	1.588 (29.40)	1.599 (28.72)	1.641 (30.58)	1.620 (27.97)	1.660 (29.15)	2.037 (28.78)	2.041 (28.86)	2.085 (31.11)	2.062 (29.21)	2.110 (31.55)
Imputation using predictive mean matching method										
10%	1.290 (5.16)	1.290 (3.90)	1.322 (5.16)	1.327 (4.76)	1.321 (2.76)	1.648 (4.17)	1.647 (4.03)	1.669 (4.93)	1.671 (4.71)	1.664 (3.71)
20%	1.360 (10.88)	1.355 (9.13)	1.385 (10.19)	1.381 (9.09)	1.402 (9.06)	1.732 (9.51)	1.730 (9.24)	1.753 (10.24)	1.746 (9.38)	1.760 (9.73)
30%	1.419 (15.64)	1.446 (16.42)	1.450 (15.40)	1.461 (15.36)	1.478 (15.00)	1.815 (14.74)	1.841 (16.29)	1.842 (15.80)	1.842 (15.43)	1.864 (16.22)
40%	1.508 (22.86)	1.550 (24.79)	1.580 (25.75)	1.572 (24.14)	1.576 (22.59)	1.937 (22.50)	1.977 (24.83)	2.006 (26.12)	1.997 (25.12)	2.001 (24.77)
Imputation using propensity−score method										
10%	1.289 (5.08)	1.291 (3.96)	1.317 (4.78)	1.326 (4.70)	1.324 (3.00)	1.647 (4.14)	1.648 (4.04)	1.664 (4.65)	1.671 (4.69)	1.669 (4.02)
20%	1.366 (11.33)	1.364 (9.84)	1.377 (9.54)	1.381 (9.06)	1.412 (9.81)	1.740 (10.03)	1.741 (9.92)	1.745 (9.69)	1.747 (9.44)	1.775 (10.65)
30%	1.439 (17.24)	1.458 (17.39)	1.464 (16.49)	1.483 (17.13)	1.495 (16.33)	1.840 (16.36)	1.859 (17.39)	1.859 (16.90)	1.873 (17.37)	1.889 (17.75)
40%	1.543 (25.76)	1.588 (27.82)	1.594 (26.84)	1.599 (26.25)	1.643 (27.83)	1.979 (25.15)	2.026 (27.92)	2.025 (27.30)	2.032 (27.32)	2.087 (30.07)

SampleSize_n250_5Treatments/table2_nonshrink_5_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	96.43	96.43	96.43	100.00	89.29	100.00	100.00	96.43	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	86.21	96.55	82.76	100.00	100.00	100.00	93.10	100.00
20%	89.66	89.66	96.55	96.55	86.21	79.31	96.55	96.55	96.55	96.55	93.10
30%	92.86	96.43	92.86	89.29	89.29	75.00	100.00	100.00	100.00	100.00	96.43
40%	89.29	89.29	96.43	96.43	89.29	71.43	100.00	100.00	96.43	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	96.55	86.21	100.00	79.31	100.00	100.00	100.00	93.10	100.00
20%	93.10	89.66	96.55	93.10	93.10	86.21	96.55	96.55	96.55	96.55	93.10
30%	92.86	92.86	92.86	78.57	85.71	64.29	96.43	100.00	96.43	96.43	96.43
40%	78.57	92.86	96.43	92.86	92.86	64.29	100.00	100.00	100.00	96.43	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	86.21	96.55	79.31	100.00	100.00	100.00	93.10	100.00
20%	96.55	89.66	96.55	96.55	89.66	86.21	96.55	96.55	96.55	96.55	93.10
30%	85.71	96.43	89.29	85.71	92.86	60.71	100.00	100.00	96.43	100.00	96.43
40%	92.86	100.00	96.43	89.29	96.43	78.57	100.00	100.00	96.43	96.43	96.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	92.86	100.00	96.43	100.00	85.71	82.14	92.86	100.00	100.00	100.00	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_5Treatments/table2_nonshrink_5_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	100.00	93.10	86.21	86.21	89.66	68.97	100.00	100.00	96.55	93.10	100.00
20%	89.66	100.00	93.10	96.55	100.00	86.21	96.55	100.00	93.10	100.00	100.00
30%	96.43	89.29	89.29	96.43	96.43	75.00	100.00	96.43	96.43	100.00	100.00
40%	96.55	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	86.21	82.76	86.21	72.41	100.00	100.00	96.55	96.55	100.00
20%	93.10	100.00	93.10	100.00	96.55	86.21	96.55	100.00	96.55	100.00	100.00
30%	92.86	82.14	89.29	96.43	96.43	67.86	100.00	89.29	96.43	100.00	100.00
40%	100.00	93.10	96.55	93.10	100.00	86.21	100.00	96.55	100.00	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	93.10	86.21	82.76	89.66	68.97	100.00	96.55	93.10	93.10	96.55
20%	89.66	100.00	93.10	100.00	96.55	82.76	96.55	100.00	96.55	100.00	100.00
30%	96.43	85.71	89.29	96.43	96.43	71.43	100.00	96.43	96.43	100.00	100.00
40%	96.55	89.66	93.10	96.55	100.00	79.31	100.00	96.55	100.00	100.00	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	96.43
Imputation using linear regression	NA
10%	93.10
20%	89.66
30%	96.43
40%	96.43
Imputation using predictive mean matching method	NA
10%	93.10
20%	89.66
30%	89.29
40%	92.86
Imputation using propensity−score method	NA
10%	93.10
20%	89.66
30%	92.86
40%	89.29
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	89.29
Imputation using linear regression	NA

MR	Joint_many
10%	93.10
20%	93.10
30%	92.86
40%	100.00
Imputation using predictive mean matching method	NA
10%	96.55
20%	93.10
30%	85.71
40%	96.55
Imputation using propensity−score method	NA
10%	89.66
20%	93.10
30%	92.86
40%	96.55

SampleSize_n250_5Treatments/table2_shrink_5_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	96.43	96.43	96.43	100.00	89.29	100.00	100.00	96.43	96.43	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	86.21	96.55	82.76	100.00	100.00	100.00	93.10	100.00
20%	89.66	89.66	96.55	96.55	86.21	79.31	96.55	96.55	96.55	96.55	93.10
30%	92.86	96.43	92.86	89.29	89.29	75.00	100.00	100.00	100.00	100.00	96.43
40%	89.29	89.29	96.43	96.43	89.29	71.43	100.00	100.00	96.43	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	96.55	86.21	100.00	79.31	100.00	100.00	100.00	93.10	100.00
20%	93.10	89.66	96.55	93.10	93.10	86.21	96.55	96.55	96.55	96.55	93.10
30%	92.86	92.86	92.86	78.57	85.71	64.29	96.43	100.00	96.43	96.43	96.43
40%	78.57	92.86	96.43	92.86	92.86	64.29	100.00	100.00	100.00	96.43	92.86
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	86.21	96.55	79.31	100.00	100.00	100.00	93.10	100.00
20%	96.55	89.66	96.55	96.55	89.66	86.21	96.55	96.55	96.55	96.55	93.10
30%	85.71	96.43	89.29	85.71	92.86	60.71	100.00	100.00	96.43	100.00	96.43
40%	92.86	100.00	96.43	89.29	96.43	78.57	100.00	100.00	96.43	96.43	96.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	92.86	100.00	96.43	100.00	85.71	82.14	96.43	100.00	96.43	100.00	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_5Treatments/table2_shrink_5_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	100.00	93.10	86.21	86.21	89.66	68.97	100.00	100.00	96.55	93.10	100.00
20%	89.66	100.00	93.10	96.55	100.00	86.21	96.55	100.00	93.10	100.00	100.00
30%	96.43	89.29	89.29	96.43	96.43	75.00	100.00	96.43	96.43	100.00	100.00
40%	96.55	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	86.21	82.76	86.21	72.41	100.00	100.00	96.55	96.55	100.00
20%	93.10	100.00	93.10	100.00	96.55	86.21	96.55	100.00	96.55	100.00	100.00
30%	92.86	82.14	89.29	96.43	96.43	67.86	100.00	89.29	96.43	100.00	100.00
40%	100.00	93.10	96.55	93.10	100.00	86.21	100.00	96.55	100.00	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	93.10	86.21	82.76	89.66	68.97	100.00	96.55	93.10	93.10	96.55
20%	89.66	100.00	93.10	100.00	96.55	82.76	96.55	100.00	96.55	100.00	100.00
30%	96.43	85.71	89.29	96.43	96.43	71.43	100.00	96.43	96.43	100.00	100.00
40%	96.55	89.66	93.10	96.55	100.00	79.31	100.00	96.55	100.00	100.00	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	92.86
Imputation using linear regression	NA
10%	93.10
20%	89.66
30%	96.43
40%	96.43
Imputation using predictive mean matching method	NA
10%	93.10
20%	89.66
30%	89.29
40%	89.29
Imputation using propensity−score method	NA
10%	93.10
20%	89.66
30%	92.86
40%	89.29
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	89.29
Imputation using linear regression	NA

MR	Joint_many
10%	93.10
20%	93.10
30%	92.86
40%	100.00
Imputation using predictive mean matching method	NA
10%	96.55
20%	93.10
30%	85.71
40%	96.55
Imputation using propensity−score method	NA
10%	89.66
20%	93.10
30%	92.86
40%	96.55

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.111	1.125	1.136	1.137	1.147	1.417	1.435	1.448	1.450	1.463
Imputation using linear regression										
10%	1.156 (3.99)	1.165 (3.47)	1.185 (4.35)	1.183 (4.00)	1.211 (5.57)	1.477 (4.18)	1.488 (3.66)	1.514 (4.54)	1.511 (4.19)	1.547 (5.76)
20%	1.230 (10.66)	1.220 (8.40)	1.242 (9.35)	1.248 (9.72)	1.259 (9.76)	1.572 (10.88)	1.559 (8.63)	1.587 (9.55)	1.595 (9.96)	1.608 (9.98)
30%	1.317 (18.46)	1.336 (18.74)	1.363 (19.96)	1.372 (20.61)	1.386 (20.83)	1.681 (18.59)	1.706 (18.85)	1.739 (20.09)	1.751 (20.73)	1.769 (20.94)
40%	1.439 (29.48)	1.433 (27.28)	1.465 (28.99)	1.454 (27.84)	1.528 (33.26)	1.838 (29.65)	1.830 (27.48)	1.871 (29.18)	1.857 (28.05)	1.950 (33.35)
Imputation using predictive mean matching method										
10%	1.144 (2.97)	1.157 (2.80)	1.177 (3.66)	1.175 (3.32)	1.203 (4.89)	1.462 (3.18)	1.478 (3.01)	1.504 (3.87)	1.501 (3.53)	1.537 (5.10)
20%	1.217 (9.49)	1.218 (8.21)	1.221 (7.54)	1.240 (9.05)	1.240 (8.14)	1.554 (9.67)	1.555 (8.37)	1.560 (7.70)	1.584 (9.23)	1.584 (8.32)
30%	1.283 (15.40)	1.302 (15.68)	1.315 (15.74)	1.315 (15.66)	1.325 (15.57)	1.640 (15.68)	1.664 (15.95)	1.681 (16.05)	1.682 (15.99)	1.695 (15.88)
40%	1.398 (25.76)	1.407 (25.01)	1.418 (24.86)	1.423 (25.10)	1.468 (27.97)	1.779 (25.55)	1.791 (24.79)	1.806 (24.66)	1.811 (24.91)	1.868 (27.71)
Imputation using propensity−score method										
10%	1.149 (3.40)	1.162 (3.20)	1.183 (4.18)	1.182 (3.93)	1.207 (5.22)	1.468 (3.59)	1.484 (3.40)	1.512 (4.38)	1.510 (4.13)	1.542 (5.42)
20%	1.218 (9.59)	1.223 (8.69)	1.223 (7.64)	1.245 (9.50)	1.242 (8.30)	1.557 (9.84)	1.563 (8.93)	1.562 (7.87)	1.592 (9.75)	1.588 (8.55)
30%	1.287 (15.82)	1.304 (15.84)	1.343 (18.24)	1.333 (17.26)	1.360 (18.62)	1.644 (16.02)	1.665 (16.02)	1.715 (18.43)	1.703 (17.47)	1.737 (18.80)
40%	1.422 (27.95)	1.458 (29.51)	1.474 (29.80)	1.457 (28.11)	1.495 (30.34)	1.815 (28.04)	1.859 (29.55)	1.880 (29.82)	1.859 (28.18)	1.908 (30.43)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.098	1.111	1.126	1.136	1.145	1.401	1.419	1.437	1.450	1.462
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.163 (6.00)	1.168 (5.08)	1.176 (4.50)	1.191 (4.90)	1.205 (5.24)	1.485 (5.99)	1.491 (5.07)	1.502 (4.49)	1.521 (4.90)	1.538 (5.22)
20%	1.214 (10.63)	1.238 (11.38)	1.248 (10.90)	1.262 (11.11)	1.280 (11.81)	1.551 (10.67)	1.581 (11.42)	1.594 (10.95)	1.611 (11.15)	1.635 (11.84)
30%	1.302 (18.58)	1.303 (17.27)	1.317 (16.96)	1.339 (17.89)	1.363 (19.07)	1.664 (18.77)	1.667 (17.47)	1.684 (17.15)	1.712 (18.07)	1.743 (19.24)
40%	1.428 (30.10)	1.438 (29.35)	1.429 (26.90)	1.467 (29.14)	1.454 (26.97)	1.826 (30.33)	1.838 (29.57)	1.827 (27.15)	1.875 (29.35)	1.860 (27.25)
Imputation using predictive mean matching method										
10%	1.159 (5.61)	1.167 (5.03)	1.173 (4.19)	1.185 (4.35)	1.197 (4.52)	1.480 (5.62)	1.490 (5.03)	1.497 (4.20)	1.513 (4.36)	1.528 (4.52)
20%	1.209 (10.13)	1.224 (10.12)	1.242 (10.33)	1.255 (10.48)	1.263 (10.31)	1.543 (10.14)	1.563 (10.15)	1.586 (10.35)	1.602 (10.49)	1.613 (10.33)
30%	1.269 (15.65)	1.274 (14.65)	1.279 (13.63)	1.314 (15.67)	1.338 (16.82)	1.623 (15.80)	1.629 (14.83)	1.636 (13.82)	1.679 (15.82)	1.710 (16.97)
40%	1.391 (26.69)	1.390 (25.05)	1.369 (21.59)	1.404 (23.60)	1.436 (25.38)	1.778 (26.87)	1.777 (25.26)	1.751 (21.84)	1.795 (23.83)	1.836 (25.58)
Imputation using propensity−score method										
10%	1.158 (5.53)	1.159 (4.33)	1.168 (3.80)	1.187 (4.54)	1.196 (4.48)	1.479 (5.55)	1.480 (4.35)	1.492 (3.81)	1.516 (4.55)	1.528 (4.50)
20%	1.216 (10.79)	1.233 (10.96)	1.248 (10.85)	1.257 (10.70)	1.277 (11.56)	1.552 (10.77)	1.574 (10.95)	1.593 (10.84)	1.605 (10.68)	1.631 (11.54)
30%	1.280 (16.63)	1.291 (16.19)	1.315 (16.80)	1.325 (16.66)	1.355 (18.30)	1.637 (16.85)	1.651 (16.40)	1.681 (16.99)	1.694 (16.86)	1.732 (18.49)
40%	1.418 (29.20)	1.399 (25.89)	1.404 (24.76)	1.453 (27.90)	1.460 (27.51)	1.813 (29.40)	1.790 (26.17)	1.797 (25.02)	1.857 (28.11)	1.867 (27.75)

SampleSize_n250_5Treatments/table3_shrink_5_treatments_MAR_n250

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.111	1.125	1.136	1.137	1.147	1.421	1.425	1.433	1.432	1.440
Imputation using linear regression										
10%	1.156 (3.99)	1.165 (3.47)	1.185 (4.35)	1.183 (4.00)	1.211 (5.57)	1.483 (4.43)	1.490 (4.56)	1.500 (4.69)	1.500 (4.74)	1.524 (5.85)
20%	1.230 (10.66)	1.220 (8.40)	1.242 (9.35)	1.248 (9.72)	1.259 (9.76)	1.569 (10.46)	1.557 (9.29)	1.583 (10.45)	1.574 (9.95)	1.589 (10.39)
30%	1.317 (18.46)	1.336 (18.74)	1.363 (19.96)	1.372 (20.61)	1.386 (20.83)	1.686 (18.70)	1.707 (19.76)	1.729 (20.63)	1.736 (21.26)	1.752 (21.68)
40%	1.439 (29.48)	1.433 (27.28)	1.465 (28.99)	1.454 (27.84)	1.528 (33.26)	1.840 (29.49)	1.828 (28.28)	1.864 (30.05)	1.850 (29.17)	1.937 (34.52)
Imputation using predictive mean matching method										
10%	1.144 (2.97)	1.157 (2.80)	1.177 (3.66)	1.175 (3.32)	1.203 (4.89)	1.469 (3.43)	1.481 (3.91)	1.491 (4.05)	1.491 (4.11)	1.515 (5.25)
20%	1.217 (9.49)	1.218 (8.21)	1.221 (7.54)	1.240 (9.05)	1.240 (8.14)	1.551 (9.20)	1.553 (8.95)	1.556 (8.61)	1.563 (9.15)	1.567 (8.83)
30%	1.283 (15.40)	1.302 (15.68)	1.315 (15.74)	1.315 (15.66)	1.325 (15.57)	1.646 (15.89)	1.664 (16.80)	1.670 (16.54)	1.667 (16.45)	1.677 (16.47)
40%	1.398 (25.76)	1.407 (25.01)	1.418 (24.86)	1.423 (25.10)	1.468 (27.97)	1.779 (25.26)	1.789 (25.53)	1.800 (25.62)	1.803 (25.89)	1.852 (28.62)
Imputation using propensity−score method										
10%	1.149 (3.40)	1.162 (3.20)	1.183 (4.18)	1.182 (3.93)	1.207 (5.22)	1.475 (3.84)	1.486 (4.31)	1.499 (4.59)	1.499 (4.68)	1.520 (5.56)
20%	1.218 (9.59)	1.223 (8.69)	1.223 (7.64)	1.245 (9.50)	1.242 (8.30)	1.555 (9.46)	1.560 (9.47)	1.559 (8.82)	1.570 (9.67)	1.569 (9.02)
30%	1.287 (15.82)	1.304 (15.84)	1.343 (18.24)	1.333 (17.26)	1.360 (18.62)	1.651 (16.19)	1.666 (16.91)	1.705 (18.99)	1.691 (18.07)	1.720 (19.50)
40%	1.422 (27.95)	1.458 (29.51)	1.474 (29.80)	1.457 (28.11)	1.495 (30.34)	1.816 (27.85)	1.857 (30.30)	1.875 (30.85)	1.852 (29.33)	1.893 (31.51)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.098	1.111	1.126	1.136	1.145	1.414	1.418	1.424	1.429	1.429
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.163 (6.00)	1.168 (5.08)	1.176 (4.50)	1.191 (4.90)	1.205 (5.24)	1.486 (5.09)	1.484 (4.66)	1.492 (4.77)	1.504 (5.25)	1.514 (5.94)
20%	1.214 (10.63)	1.238 (11.38)	1.248 (10.90)	1.262 (11.11)	1.280 (11.81)	1.560 (10.31)	1.578 (11.28)	1.585 (11.28)	1.600 (11.95)	1.614 (12.93)
30%	1.302 (18.58)	1.303 (17.27)	1.317 (16.96)	1.339 (17.89)	1.363 (19.07)	1.666 (17.85)	1.665 (17.39)	1.678 (17.85)	1.701 (19.02)	1.723 (20.55)
40%	1.428 (30.10)	1.438 (29.35)	1.429 (26.90)	1.467 (29.14)	1.454 (26.97)	1.829 (29.36)	1.837 (29.49)	1.825 (28.18)	1.868 (30.66)	1.850 (29.43)
Imputation using predictive mean matching method										
10%	1.159 (5.61)	1.167 (5.03)	1.173 (4.19)	1.185 (4.35)	1.197 (4.52)	1.481 (4.72)	1.483 (4.58)	1.488 (4.46)	1.497 (4.72)	1.504 (5.26)
20%	1.209 (10.13)	1.224 (10.12)	1.242 (10.33)	1.255 (10.48)	1.263 (10.31)	1.552 (9.76)	1.561 (10.03)	1.576 (10.69)	1.590 (11.27)	1.591 (11.34)
30%	1.269 (15.65)	1.274 (14.65)	1.279 (13.63)	1.314 (15.67)	1.338 (16.82)	1.626 (15.03)	1.629 (14.82)	1.630 (14.45)	1.668 (16.71)	1.688 (18.14)
40%	1.391 (26.69)	1.390 (25.05)	1.369 (21.59)	1.404 (23.60)	1.436 (25.38)	1.783 (26.11)	1.777 (25.27)	1.752 (23.05)	1.788 (25.13)	1.826 (27.75)
Imputation using propensity−score method										
10%	1.158 (5.53)	1.159 (4.33)	1.168 (3.80)	1.187 (4.54)	1.196 (4.48)	1.480 (4.68)	1.474 (3.94)	1.483 (4.14)	1.499 (4.89)	1.504 (5.23)
20%	1.216 (10.79)	1.233 (10.96)	1.248 (10.85)	1.257 (10.70)	1.277 (11.56)	1.560 (10.36)	1.572 (10.82)	1.584 (11.23)	1.593 (11.48)	1.609 (12.62)
30%	1.280 (16.63)	1.291 (16.19)	1.315 (16.80)	1.325 (16.66)	1.355 (18.30)	1.641 (16.03)	1.651 (16.37)	1.676 (17.69)	1.684 (17.85)	1.712 (19.77)
40%	1.418 (29.20)	1.399 (25.89)	1.404 (24.76)	1.453 (27.90)	1.460 (27.51)	1.817 (28.52)	1.790 (26.19)	1.797 (26.19)	1.850 (29.41)	1.858 (29.98)

SampleSize_n300_5Treatments/table2_nonshrink_5_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	96.43	100.00	92.86	92.86	82.14	100.00	100.00	100.00	92.86	100
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	100.00	100.00	96.55	96.55	93.10	100.00	100.00	100.00	100.00	100
30%	89.29	96.43	96.43	96.43	96.43	78.57	96.43	100.00	100.00	100.00	100
40%	89.66	96.55	96.55	96.55	96.55	75.86	96.55	100.00	100.00	100.00	100
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	96.55	96.55	96.55	100.00	86.21	100.00	100.00	100.00	96.55	100
30%	89.29	92.86	96.43	92.86	92.86	71.43	92.86	100.00	96.43	100.00	100
40%	86.21	86.21	93.10	96.55	100.00	68.97	96.55	96.55	100.00	100.00	100
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	96.55	96.55	96.55	100.00	86.21	100.00	100.00	100.00	100.00	100
30%	92.86	96.43	92.86	92.86	92.86	78.57	96.43	100.00	96.43	100.00	100
40%	89.66	96.55	96.55	96.55	100.00	79.31	96.55	96.55	100.00	100.00	100
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	100.00	96.55	93.10	96.55	82.76	100.00	100.00	100.00	96.55	100
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_5Treatments/table2_nonshrink_5_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	86.21	93.10	96.55	89.66	86.21	79.31	93.10	96.55	96.55	96.55	96.55
20%	85.71	89.29	85.71	89.29	92.86	67.86	92.86	92.86	92.86	96.43	96.43
30%	96.43	100.00	96.43	96.43	96.43	85.71	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	100.00	89.66	86.21	100.00	100.00	100.00	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	89.66	96.55	89.66	86.21	79.31	93.10	96.55	96.55	96.55	96.55
20%	82.14	89.29	85.71	89.29	92.86	60.71	92.86	92.86	92.86	96.43	96.43
30%	96.43	100.00	96.43	96.43	96.43	85.71	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	96.55	89.66	82.76	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	82.76	79.31	93.10	96.55	96.55	96.55	93.10
20%	82.14	92.86	85.71	89.29	92.86	60.71	92.86	92.86	92.86	96.43	96.43
30%	96.43	96.43	96.43	96.43	92.86	82.14	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	100.00	89.66	86.21	100.00	100.00	100.00	100.00	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	92.86
Imputation using linear regression	NA
10%	100.00
20%	100.00
30%	96.43
40%	96.55
Imputation using predictive mean matching method	NA
10%	100.00
20%	96.55
30%	92.86
40%	93.10
Imputation using propensity−score method	NA
10%	100.00
20%	100.00
30%	96.43
40%	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	82.14
30%	96.43
40%	96.55
Imputation using predictive mean matching method	NA
10%	89.66
20%	82.14
30%	96.43
40%	96.55
Imputation using propensity−score method	NA
10%	86.21
20%	82.14
30%	96.43
40%	100.00

SampleSize_n300_5Treatments/table2_shrink_5_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	96.43	100.00	92.86	92.86	82.14	100.00	100.00	100.00	92.86	100
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	100.00	100.00	96.55	96.55	93.10	100.00	100.00	100.00	100.00	100
30%	89.29	96.43	96.43	96.43	96.43	78.57	96.43	100.00	100.00	100.00	100
40%	89.66	96.55	96.55	96.55	96.55	75.86	96.55	100.00	100.00	100.00	100
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	96.55	96.55	96.55	100.00	86.21	100.00	100.00	100.00	96.55	100
30%	89.29	92.86	96.43	92.86	92.86	71.43	92.86	100.00	96.43	100.00	100
40%	86.21	86.21	93.10	96.55	100.00	68.97	96.55	96.55	100.00	100.00	100
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	96.55	96.55	96.55	100.00	86.21	100.00	100.00	100.00	100.00	100
30%	92.86	96.43	92.86	92.86	92.86	78.57	96.43	100.00	96.43	100.00	100
40%	89.66	96.55	96.55	96.55	100.00	79.31	96.55	96.55	96.55	100.00	100
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	100.00	96.55	93.10	96.55	82.76	100.00	100.00	100.00	96.55	100
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_5Treatments/table2_shrink_5_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	86.21	93.10	96.55	89.66	86.21	79.31	93.10	96.55	96.55	96.55	96.55
20%	85.71	89.29	85.71	89.29	92.86	67.86	92.86	92.86	92.86	96.43	96.43
30%	96.43	100.00	96.43	96.43	96.43	85.71	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	100.00	89.66	86.21	100.00	100.00	100.00	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	89.66	96.55	89.66	86.21	79.31	93.10	96.55	96.55	96.55	96.55
20%	82.14	89.29	85.71	89.29	92.86	60.71	92.86	92.86	92.86	96.43	96.43
30%	96.43	100.00	96.43	96.43	96.43	85.71	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	96.55	89.66	82.76	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	82.76	79.31	93.10	96.55	96.55	93.10	93.10
20%	82.14	92.86	85.71	89.29	92.86	60.71	92.86	92.86	92.86	96.43	96.43
30%	96.43	96.43	96.43	96.43	92.86	82.14	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	100.00	89.66	86.21	100.00	100.00	100.00	100.00	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	92.86
Imputation using linear regression	NA
10%	100.00
20%	100.00
30%	96.43
40%	96.55
Imputation using predictive mean matching method	NA
10%	100.00
20%	96.55
30%	92.86
40%	93.10
Imputation using propensity−score method	NA
10%	100.00
20%	100.00
30%	96.43
40%	89.66
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	82.14
30%	96.43
40%	96.55
Imputation using predictive mean matching method	NA
10%	89.66
20%	82.14
30%	96.43
40%	96.55
Imputation using propensity−score method	NA
10%	86.21
20%	82.14
30%	96.43
40%	100.00

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.025	1.034	1.036	1.053	1.056	1.307	1.319	1.321	1.343	1.347
Imputation using linear regression										
10%	1.051 (2.53)	1.062 (2.72)	1.079 (4.17)	1.090 (3.53)	1.098 (3.94)	1.342 (2.66)	1.357 (2.87)	1.378 (4.32)	1.392 (3.67)	1.402 (4.08)
20%	1.105 (7.85)	1.127 (8.93)	1.129 (8.98)	1.134 (7.66)	1.164 (10.14)	1.413 (8.12)	1.440 (9.20)	1.443 (9.26)	1.449 (7.92)	1.487 (10.41)
30%	1.183 (15.40)	1.212 (17.18)	1.213 (17.14)	1.238 (17.58)	1.247 (18.00)	1.513 (15.75)	1.550 (17.49)	1.552 (17.48)	1.583 (17.89)	1.594 (18.34)
40%	1.289 (25.79)	1.305 (26.20)	1.314 (26.83)	1.337 (26.97)	1.335 (26.36)	1.650 (26.23)	1.670 (26.63)	1.681 (27.27)	1.711 (27.37)	1.708 (26.81)
Imputation using predictive mean matching method										
10%	1.045 (1.97)	1.063 (2.75)	1.073 (3.63)	1.087 (3.22)	1.094 (3.55)	1.334 (2.08)	1.357 (2.87)	1.370 (3.75)	1.388 (3.33)	1.397 (3.66)
20%	1.091 (6.44)	1.114 (7.66)	1.118 (7.92)	1.125 (6.80)	1.148 (8.71)	1.395 (6.70)	1.423 (7.91)	1.429 (8.18)	1.438 (7.05)	1.468 (8.97)
30%	1.155 (12.67)	1.160 (12.13)	1.175 (13.46)	1.196 (13.53)	1.211 (14.59)	1.477 (13.03)	1.484 (12.49)	1.503 (13.82)	1.529 (13.86)	1.548 (14.93)
40%	1.245 (21.45)	1.246 (20.42)	1.267 (22.30)	1.289 (22.43)	1.296 (22.69)	1.594 (21.97)	1.595 (20.96)	1.623 (22.83)	1.651 (22.93)	1.660 (23.21)
Imputation using propensity−score method										
10%	1.044 (1.86)	1.063 (2.77)	1.078 (4.06)	1.090 (3.46)	1.098 (3.89)	1.333 (2.00)	1.357 (2.91)	1.376 (4.19)	1.391 (3.60)	1.401 (4.02)
20%	1.100 (7.31)	1.120 (8.26)	1.120 (8.12)	1.134 (7.71)	1.156 (9.40)	1.406 (7.53)	1.431 (8.47)	1.431 (8.35)	1.449 (7.92)	1.477 (9.63)
30%	1.164 (13.58)	1.183 (14.41)	1.195 (15.36)	1.203 (14.25)	1.222 (15.65)	1.489 (13.93)	1.513 (14.73)	1.528 (15.70)	1.539 (14.58)	1.563 (15.99)
40%	1.295 (26.36)	1.290 (24.70)	1.303 (25.84)	1.325 (25.81)	1.327 (25.57)	1.655 (26.64)	1.649 (25.00)	1.666 (26.16)	1.693 (26.09)	1.696 (25.88)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.014	1.017	1.029	1.037	1.048	1.294	1.298	1.314	1.323	1.337
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.055 (4.02)	1.068 (5.02)	1.079 (4.88)	1.092 (5.38)	1.094 (4.43)	1.345 (3.93)	1.362 (4.93)	1.377 (4.79)	1.393 (5.29)	1.395 (4.34)
20%	1.112 (9.68)	1.125 (10.66)	1.135 (10.28)	1.140 (9.97)	1.149 (9.70)	1.421 (9.79)	1.438 (10.77)	1.450 (10.40)	1.457 (10.08)	1.469 (9.81)
30%	1.194 (17.71)	1.209 (18.94)	1.225 (18.98)	1.225 (18.18)	1.224 (16.86)	1.526 (17.86)	1.546 (19.09)	1.565 (19.14)	1.566 (18.35)	1.565 (17.05)
40%	1.286 (26.79)	1.296 (27.53)	1.303 (26.55)	1.308 (26.18)	1.324 (26.34)	1.644 (27.03)	1.658 (27.79)	1.666 (26.81)	1.673 (26.46)	1.693 (26.59)
Imputation using predictive mean matching method										
10%	1.050 (3.54)	1.061 (4.38)	1.076 (4.56)	1.086 (4.76)	1.089 (3.91)	1.340 (3.49)	1.354 (4.34)	1.373 (4.51)	1.386 (4.71)	1.389 (3.87)
20%	1.105 (8.94)	1.113 (9.46)	1.123 (9.10)	1.125 (8.52)	1.142 (9.01)	1.411 (8.98)	1.421 (9.50)	1.434 (9.15)	1.437 (8.57)	1.458 (9.05)
30%	1.181 (16.44)	1.189 (16.90)	1.199 (16.46)	1.215 (17.24)	1.218 (16.29)	1.507 (16.38)	1.517 (16.87)	1.530 (16.43)	1.550 (17.18)	1.555 (16.24)
40%	1.255 (23.80)	1.267 (24.61)	1.281 (24.44)	1.285 (23.99)	1.271 (21.33)	1.605 (23.98)	1.620 (24.82)	1.638 (24.65)	1.644 (24.21)	1.626 (21.60)
Imputation using propensity−score method										
10%	1.050 (3.59)	1.061 (4.38)	1.074 (4.35)	1.085 (4.67)	1.086 (3.71)	1.340 (3.55)	1.354 (4.34)	1.370 (4.31)	1.384 (4.63)	1.386 (3.66)
20%	1.110 (9.46)	1.122 (10.32)	1.139 (10.70)	1.142 (10.22)	1.156 (10.33)	1.417 (9.48)	1.432 (10.35)	1.455 (10.72)	1.459 (10.23)	1.476 (10.35)
30%	1.179 (16.28)	1.197 (17.74)	1.207 (17.24)	1.230 (18.64)	1.234 (17.76)	1.506 (16.37)	1.529 (17.85)	1.542 (17.35)	1.571 (18.71)	1.576 (17.84)
40%	1.272 (25.45)	1.288 (26.68)	1.313 (27.54)	1.292 (24.67)	1.290 (23.16)	1.625 (25.56)	1.646 (26.80)	1.677 (27.64)	1.652 (24.84)	1.650 (23.34)

SampleSize_n300_5Treatments/table3_shrink_5_treatments_MAR_n300

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.025	1.034	1.036	1.053	1.056	1.307	1.312	1.313	1.318	1.320
Imputation using linear regression										
10%	1.051 (2.53)	1.062 (2.72)	1.079 (4.17)	1.090 (3.53)	1.098 (3.94)	1.351 (3.33)	1.354 (3.22)	1.369 (4.21)	1.376 (4.38)	1.384 (4.82)
20%	1.105 (7.85)	1.127 (8.93)	1.129 (8.98)	1.134 (7.66)	1.164 (10.14)	1.419 (8.51)	1.435 (9.43)	1.436 (9.36)	1.439 (9.23)	1.466 (11.05)
30%	1.183 (15.40)	1.212 (17.18)	1.213 (17.14)	1.238 (17.58)	1.247 (18.00)	1.518 (16.14)	1.549 (18.08)	1.547 (17.80)	1.576 (19.56)	1.582 (19.80)
40%	1.289 (25.79)	1.305 (26.20)	1.314 (26.83)	1.337 (26.97)	1.335 (26.36)	1.653 (26.45)	1.670 (27.34)	1.676 (27.63)	1.704 (29.27)	1.697 (28.52)
Imputation using predictive mean matching method										
10%	1.045 (1.97)	1.063 (2.75)	1.073 (3.63)	1.087 (3.22)	1.094 (3.55)	1.343 (2.72)	1.355 (3.27)	1.363 (3.77)	1.371 (4.05)	1.378 (4.39)
20%	1.091 (6.44)	1.114 (7.66)	1.118 (7.92)	1.125 (6.80)	1.148 (8.71)	1.400 (7.04)	1.418 (8.10)	1.421 (8.18)	1.427 (8.26)	1.447 (9.61)
30%	1.155 (12.67)	1.160 (12.13)	1.175 (13.46)	1.196 (13.53)	1.211 (14.59)	1.484 (13.48)	1.485 (13.20)	1.500 (14.21)	1.522 (15.50)	1.536 (16.36)
40%	1.245 (21.45)	1.246 (20.42)	1.267 (22.30)	1.289 (22.43)	1.296 (22.69)	1.599 (22.27)	1.596 (21.72)	1.617 (23.15)	1.645 (24.79)	1.647 (24.77)
Imputation using propensity−score method										
10%	1.044 (1.86)	1.063 (2.77)	1.078 (4.06)	1.090 (3.46)	1.098 (3.89)	1.343 (2.68)	1.355 (3.31)	1.368 (4.20)	1.375 (4.36)	1.383 (4.79)
20%	1.100 (7.31)	1.120 (8.26)	1.120 (8.12)	1.134 (7.71)	1.156 (9.40)	1.410 (7.87)	1.425 (8.65)	1.424 (8.41)	1.439 (9.21)	1.456 (10.32)
30%	1.164 (13.58)	1.183 (14.41)	1.195 (15.36)	1.203 (14.25)	1.222 (15.65)	1.496 (14.39)	1.513 (15.35)	1.524 (16.02)	1.532 (16.28)	1.550 (17.44)
40%	1.295 (26.36)	1.290 (24.70)	1.303 (25.84)	1.325 (25.81)	1.327 (25.57)	1.659 (26.86)	1.649 (25.75)	1.661 (26.48)	1.687 (28.05)	1.685 (27.64)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.014	1.017	1.029	1.037	1.048	1.297	1.300	1.302	1.305	1.312
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.055 (4.02)	1.068 (5.02)	1.079 (4.88)	1.092 (5.38)	1.094 (4.43)	1.346 (3.79)	1.357 (4.43)	1.366 (4.93)	1.375 (5.34)	1.378 (5.03)
20%	1.112 (9.68)	1.125 (10.66)	1.135 (10.28)	1.140 (9.97)	1.149 (9.70)	1.424 (9.81)	1.433 (10.26)	1.442 (10.73)	1.446 (10.78)	1.449 (10.43)
30%	1.194 (17.71)	1.209 (18.94)	1.225 (18.98)	1.225 (18.18)	1.224 (16.86)	1.529 (17.93)	1.542 (18.67)	1.557 (19.59)	1.555 (19.18)	1.555 (18.47)
40%	1.286 (26.79)	1.296 (27.53)	1.303 (26.55)	1.308 (26.18)	1.324 (26.34)	1.648 (27.09)	1.656 (27.39)	1.660 (27.50)	1.664 (27.52)	1.682 (28.20)
Imputation using predictive mean matching method										
10%	1.050 (3.54)	1.061 (4.38)	1.076 (4.56)	1.086 (4.76)	1.089 (3.91)	1.340 (3.35)	1.350 (3.86)	1.362 (4.63)	1.367 (4.76)	1.372 (4.54)
20%	1.105 (8.94)	1.113 (9.46)	1.123 (9.10)	1.125 (8.52)	1.142 (9.01)	1.413 (8.97)	1.417 (8.99)	1.425 (9.44)	1.426 (9.24)	1.439 (9.67)
30%	1.181 (16.44)	1.189 (16.90)	1.199 (16.46)	1.215 (17.24)	1.218 (16.29)	1.511 (16.53)	1.513 (16.39)	1.522 (16.88)	1.540 (18.00)	1.546 (17.78)
40%	1.255 (23.80)	1.267 (24.61)	1.281 (24.44)	1.285 (23.99)	1.271 (21.33)	1.609 (24.08)	1.619 (24.54)	1.631 (25.30)	1.635 (25.26)	1.617 (23.25)
Imputation using propensity−score method										
10%	1.050 (3.59)	1.061 (4.38)	1.074 (4.35)	1.085 (4.67)	1.086 (3.71)	1.341 (3.41)	1.350 (3.91)	1.360 (4.47)	1.366 (4.67)	1.369 (4.32)
20%	1.110 (9.46)	1.122 (10.32)	1.139 (10.70)	1.142 (10.22)	1.156 (10.33)	1.419 (9.45)	1.427 (9.78)	1.445 (10.99)	1.448 (10.96)	1.457 (11.03)
30%	1.179 (16.28)	1.197 (17.74)	1.207 (17.24)	1.230 (18.64)	1.234 (17.76)	1.510 (16.44)	1.526 (17.42)	1.534 (17.84)	1.560 (19.55)	1.566 (19.37)
40%	1.272 (25.45)	1.288 (26.68)	1.313 (27.54)	1.292 (24.67)	1.290 (23.16)	1.629 (25.62)	1.643 (26.41)	1.670 (28.30)	1.643 (25.92)	1.639 (24.91)

SampleSize_n50_5Treatments/table2_nonshrink_5_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	93.10	93.10	100.00	100.00	82.76	96.55	96.55	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	82.76	89.66	93.10	96.55	93.10	68.97	93.10	96.55	100.00	96.55	100.00
20%	100.00	96.43	96.43	92.86	96.43	85.71	100.00	96.43	100.00	100.00	100.00
30%	100.00	93.10	89.66	93.10	93.10	72.41	100.00	96.55	100.00	100.00	96.55
40%	96.55	100.00	96.55	100.00	93.10	86.21	100.00	100.00	100.00	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	89.66	93.10	96.55	93.10	72.41	93.10	96.55	96.55	100.00	100.00
20%	100.00	96.43	96.43	92.86	96.43	85.71	100.00	96.43	100.00	100.00	100.00
30%	100.00	96.55	86.21	93.10	82.76	68.97	100.00	96.55	93.10	100.00	93.10
40%	93.10	96.55	89.66	93.10	89.66	68.97	100.00	100.00	100.00	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	93.10	96.55	93.10	72.41	93.10	96.55	96.55	96.55	100.00
20%	96.43	96.43	96.43	92.86	96.43	82.14	100.00	96.43	100.00	96.43	100.00
30%	96.55	89.66	86.21	93.10	86.21	58.62	100.00	96.55	96.55	100.00	93.10
40%	93.10	96.55	89.66	96.55	89.66	79.31	100.00	100.00	100.00	100.00	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.66	93.10	96.55	100.00	93.10	82.76	96.55	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_5Treatments/table2_nonshrink_5_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.43	100.00	96.43	100.00	96.43	89.29	100.00	100.00	100.00	100.00	100.00
20%	96.43	100.00	89.29	96.43	96.43	85.71	96.43	100.00	96.43	100.00	100.00
30%	89.66	96.55	93.10	96.55	96.55	82.76	96.55	100.00	100.00	96.55	100.00
40%	86.21	89.66	100.00	89.66	96.55	79.31	100.00	96.55	100.00	93.10	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	100.00	96.43	85.71	100.00	100.00	100.00	100.00	100.00
20%	96.43	100.00	85.71	96.43	96.43	82.14	96.43	100.00	96.43	100.00	100.00
30%	89.66	96.55	93.10	96.55	82.76	72.41	96.55	96.55	100.00	96.55	100.00
40%	82.76	86.21	93.10	82.76	93.10	79.31	93.10	89.66	100.00	89.66	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	92.86	100.00	96.43	82.14	100.00	100.00	100.00	100.00	100.00
20%	96.43	96.43	89.29	96.43	96.43	85.71	96.43	100.00	92.86	100.00	100.00
30%	89.66	96.55	93.10	96.55	86.21	75.86	96.55	96.55	100.00	96.55	100.00
40%	82.76	89.66	100.00	89.66	89.66	82.76	93.10	96.55	100.00	93.10	96.55

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.10
Imputation using linear regression	NA
10%	86.21
20%	96.43
30%	93.10
40%	96.55
Imputation using predictive mean matching method	NA
10%	86.21
20%	96.43
30%	82.76
40%	100.00
Imputation using propensity−score method	NA
10%	82.76
20%	92.86
30%	86.21
40%	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	100.00
20%	92.86
30%	96.55
40%	89.66
Imputation using predictive mean matching method	NA
10%	100.00
20%	92.86
30%	93.10
40%	89.66
Imputation using propensity−score method	NA
10%	100.00
20%	89.29
30%	93.10
40%	89.66

SampleSize_n50_5Treatments/table2_shrink_5_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	93.10	93.10	100.00	100.00	82.76	96.55	96.55	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	82.76	89.66	93.10	96.55	93.10	68.97	96.55	96.55	100.00	100.00	100.00
20%	100.00	96.43	96.43	92.86	96.43	85.71	100.00	96.43	100.00	100.00	100.00
30%	100.00	93.10	89.66	93.10	93.10	72.41	100.00	96.55	93.10	100.00	96.55
40%	96.55	100.00	96.55	100.00	93.10	86.21	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	89.66	93.10	96.55	93.10	72.41	93.10	96.55	96.55	100.00	100.00
20%	100.00	96.43	96.43	92.86	96.43	85.71	100.00	96.43	100.00	100.00	100.00
30%	100.00	96.55	86.21	93.10	82.76	68.97	100.00	96.55	93.10	100.00	93.10
40%	93.10	96.55	89.66	93.10	89.66	68.97	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	93.10	96.55	93.10	72.41	93.10	96.55	96.55	100.00	100.00
20%	96.43	96.43	96.43	92.86	96.43	82.14	100.00	96.43	100.00	96.43	100.00
30%	96.55	89.66	86.21	93.10	86.21	58.62	100.00	96.55	96.55	100.00	93.10
40%	93.10	96.55	89.66	96.55	89.66	79.31	100.00	100.00	100.00	100.00	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.66	93.10	96.55	100.00	93.10	82.76	96.55	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_5Treatments/table2_shrink_5_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.43	100.00	96.43	100.00	96.43	89.29	100.00	100.00	100.00	100.00	100.00
20%	96.43	100.00	89.29	96.43	96.43	85.71	96.43	100.00	96.43	100.00	100.00
30%	89.66	96.55	93.10	96.55	96.55	82.76	96.55	100.00	100.00	96.55	100.00
40%	86.21	89.66	100.00	89.66	96.55	79.31	100.00	96.55	100.00	93.10	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	100.00	96.43	85.71	100.00	100.00	100.00	100.00	100.00
20%	96.43	100.00	85.71	96.43	96.43	82.14	96.43	100.00	96.43	100.00	100.00
30%	89.66	96.55	93.10	96.55	82.76	72.41	96.55	96.55	100.00	96.55	100.00
40%	82.76	86.21	93.10	82.76	93.10	79.31	96.55	93.10	100.00	89.66	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	92.86	100.00	96.43	82.14	100.00	100.00	100.00	100.00	100.00
20%	96.43	96.43	89.29	96.43	96.43	85.71	96.43	100.00	96.43	100.00	100.00
30%	89.66	96.55	93.10	96.55	86.21	75.86	93.10	96.55	96.55	96.55	100.00
40%	82.76	89.66	100.00	89.66	89.66	82.76	96.55	96.55	100.00	93.10	96.55

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.10
Imputation using linear regression	NA
10%	93.10
20%	96.43
30%	86.21
40%	100.00
Imputation using predictive mean matching method	NA
10%	86.21
20%	96.43
30%	82.76
40%	96.55
Imputation using propensity−score method	NA
10%	86.21
20%	92.86
30%	86.21
40%	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	100.00
20%	92.86
30%	96.55
40%	89.66
Imputation using predictive mean matching method	NA
10%	100.00
20%	92.86
30%	93.10
40%	89.66
Imputation using propensity−score method	NA
10%	100.00
20%	92.86
30%	89.66
40%	89.66

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	2.519	2.502	2.547	2.551	2.593	3.199	3.178	3.235	3.240	3.293
Imputation using linear regression										
10%	2.628 (4.34)	2.621 (4.77)	2.654 (4.20)	2.719 (6.58)	2.710 (4.50)	3.356 (4.90)	3.347 (5.32)	3.389 (4.75)	3.471 (7.13)	3.460 (5.06)
20%	2.836 (12.58)	2.819 (12.70)	2.906 (14.09)	2.888 (13.22)	2.914 (12.37)	3.618 (13.07)	3.597 (13.20)	3.707 (14.59)	3.685 (13.73)	3.718 (12.89)
30%	2.872 (14.02)	3.000 (19.90)	2.981 (17.03)	3.089 (21.08)	3.038 (17.19)	3.672 (14.76)	3.834 (20.67)	3.811 (17.79)	3.950 (21.92)	3.885 (17.95)
40%	3.358 (33.29)	3.261 (30.35)	3.255 (27.79)	3.431 (34.48)	3.412 (31.59)	4.301 (34.42)	4.181 (31.58)	4.173 (28.98)	4.397 (35.69)	4.370 (32.70)
Imputation using predictive mean matching method										
10%	2.573 (2.16)	2.600 (3.94)	2.612 (2.56)	2.674 (4.83)	2.661 (2.63)	3.286 (2.69)	3.320 (4.47)	3.335 (3.08)	3.414 (5.35)	3.397 (3.15)
20%	2.721 (8.04)	2.699 (7.87)	2.820 (10.72)	2.792 (9.47)	2.806 (8.23)	3.471 (8.48)	3.442 (8.33)	3.596 (11.16)	3.562 (9.93)	3.580 (8.71)
30%	2.683 (6.52)	2.784 (11.28)	2.802 (10.00)	2.884 (13.08)	2.811 (8.41)	3.429 (7.19)	3.559 (12.01)	3.582 (10.72)	3.689 (13.87)	3.595 (9.16)
40%	2.982 (18.39)	2.964 (18.49)	3.017 (18.45)	3.020 (18.39)	3.043 (17.35)	3.815 (19.23)	3.794 (19.39)	3.859 (19.27)	3.866 (19.32)	3.892 (18.19)
Imputation using propensity−score method										
10%	2.586 (2.65)	2.591 (3.55)	2.618 (2.78)	2.678 (5.00)	2.664 (2.74)	3.302 (3.22)	3.309 (4.13)	3.344 (3.35)	3.420 (5.56)	3.402 (3.30)
20%	2.746 (9.02)	2.707 (8.22)	2.815 (10.52)	2.800 (9.77)	2.838 (9.44)	3.502 (9.47)	3.454 (8.69)	3.591 (10.99)	3.572 (10.23)	3.620 (9.91)
30%	2.679 (6.35)	2.811 (12.35)	2.849 (11.85)	2.951 (15.70)	2.855 (10.10)	3.428 (7.14)	3.596 (13.17)	3.644 (12.64)	3.776 (16.55)	3.653 (10.93)
40%	3.041 (20.74)	3.034 (21.29)	3.063 (20.27)	3.166 (24.12)	3.190 (23.04)	3.890 (21.59)	3.882 (22.16)	3.918 (21.11)	4.051 (25.02)	4.079 (23.86)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	2.477	2.539	2.550	2.584	2.582	3.134	3.213	3.227	3.269	3.267
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	2.642 (6.65)	2.702 (6.43)	2.715 (6.47)	2.729 (5.59)	2.749 (6.49)	3.376 (7.72)	3.453 (7.49)	3.470 (7.52)	3.486 (6.63)	3.513 (7.55)
20%	2.785 (12.44)	2.858 (12.56)	2.759 (8.18)	2.853 (10.39)	2.885 (11.73)	3.545 (13.12)	3.638 (13.25)	3.513 (8.86)	3.632 (11.08)	3.673 (12.45)
30%	2.902 (17.15)	3.011 (18.57)	3.018 (18.32)	3.003 (16.22)	3.074 (19.07)	3.703 (18.18)	3.841 (19.57)	3.852 (19.38)	3.832 (17.20)	3.923 (20.09)
40%	3.258 (31.53)	3.331 (31.19)	3.418 (34.03)	3.494 (35.22)	3.409 (32.03)	4.158 (32.69)	4.250 (32.28)	4.362 (35.18)	4.460 (36.41)	4.350 (33.15)
Imputation using predictive mean matching method										
10%	2.587 (4.45)	2.654 (4.50)	2.652 (3.98)	2.696 (4.34)	2.676 (3.64)	3.306 (5.49)	3.391 (5.54)	3.388 (5.00)	3.445 (5.35)	3.419 (4.68)
20%	2.656 (7.21)	2.736 (7.75)	2.694 (5.61)	2.783 (7.71)	2.791 (8.09)	3.380 (7.88)	3.482 (8.39)	3.428 (6.23)	3.543 (8.35)	3.552 (8.74)
30%	2.709 (9.34)	2.769 (9.06)	2.809 (10.14)	2.759 (6.77)	2.859 (10.75)	3.454 (10.24)	3.532 (9.94)	3.584 (11.07)	3.520 (7.65)	3.647 (11.65)
40%	2.943 (18.80)	2.988 (17.68)	3.024 (18.55)	3.194 (23.61)	3.018 (16.89)	3.756 (19.87)	3.815 (18.74)	3.862 (19.67)	4.075 (24.65)	3.853 (17.95)
Imputation using propensity−score method										
10%	2.611 (5.41)	2.669 (5.10)	2.677 (4.95)	2.696 (4.33)	2.704 (4.73)	3.336 (6.46)	3.410 (6.14)	3.420 (5.98)	3.444 (5.35)	3.455 (5.77)
20%	2.718 (9.73)	2.760 (8.71)	2.701 (5.90)	2.820 (9.12)	2.794 (8.22)	3.460 (10.43)	3.516 (9.43)	3.439 (6.58)	3.591 (9.82)	3.559 (8.96)
30%	2.747 (10.88)	2.802 (10.34)	2.880 (12.91)	2.823 (9.26)	2.942 (13.96)	3.504 (11.83)	3.575 (11.27)	3.674 (13.85)	3.602 (10.17)	3.752 (14.87)
40%	2.981 (20.34)	3.019 (18.89)	3.069 (20.32)	3.177 (22.95)	3.033 (17.46)	3.813 (21.68)	3.862 (20.22)	3.926 (21.67)	4.065 (24.32)	3.881 (18.80)

SampleSize_n50_5Treatments/table3_shrink_5_treatments_MAR_n50

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	2.519	2.502	2.547	2.551	2.593	3.226	3.213	3.226	3.226	3.245
Imputation using linear regression										
10%	2.628 (4.34)	2.621 (4.77)	2.654 (4.20)	2.719 (6.58)	2.710 (4.50)	3.357 (4.06)	3.367 (4.80)	3.348 (3.79)	3.419 (5.99)	3.388 (4.40)
20%	2.836 (12.58)	2.819 (12.70)	2.906 (14.09)	2.888 (13.22)	2.914 (12.37)	3.624 (12.33)	3.594 (11.86)	3.644 (12.96)	3.627 (12.40)	3.659 (12.75)
30%	2.872 (14.02)	3.000 (19.90)	2.981 (17.03)	3.089 (21.08)	3.038 (17.19)	3.720 (15.30)	3.829 (19.17)	3.808 (18.05)	3.920 (21.51)	3.867 (19.18)
40%	3.358 (33.29)	3.261 (30.35)	3.255 (27.79)	3.431 (34.48)	3.412 (31.59)	4.320 (33.91)	4.189 (30.38)	4.178 (29.52)	4.365 (35.31)	4.356 (34.25)
Imputation using predictive mean matching method										
10%	2.573 (2.16)	2.600 (3.94)	2.612 (2.56)	2.674 (4.83)	2.661 (2.63)	3.289 (1.95)	3.330 (3.64)	3.293 (2.07)	3.352 (3.88)	3.320 (2.32)
20%	2.721 (8.04)	2.699 (7.87)	2.820 (10.72)	2.792 (9.47)	2.806 (8.23)	3.477 (7.78)	3.439 (7.04)	3.530 (9.43)	3.493 (8.27)	3.524 (8.59)
30%	2.683 (6.52)	2.784 (11.28)	2.802 (10.00)	2.884 (13.08)	2.811 (8.41)	3.482 (7.92)	3.556 (10.68)	3.577 (10.87)	3.653 (13.23)	3.578 (10.26)
40%	2.982 (18.39)	2.964 (18.49)	3.017 (18.45)	3.020 (18.39)	3.043 (17.35)	3.843 (19.12)	3.805 (18.42)	3.862 (19.73)	3.841 (19.06)	3.873 (19.35)
Imputation using propensity−score method										
10%	2.586 (2.65)	2.591 (3.55)	2.618 (2.78)	2.678 (5.00)	2.664 (2.74)	3.307 (2.50)	3.325 (3.47)	3.304 (2.41)	3.362 (4.20)	3.328 (2.57)
20%	2.746 (9.02)	2.707 (8.22)	2.815 (10.52)	2.800 (9.77)	2.838 (9.44)	3.505 (8.62)	3.448 (7.32)	3.523 (9.21)	3.507 (8.71)	3.563 (9.79)
30%	2.679 (6.35)	2.811 (12.35)	2.849 (11.85)	2.951 (15.70)	2.855 (10.10)	3.482 (7.91)	3.595 (11.89)	3.643 (12.93)	3.745 (16.07)	3.637 (12.09)
40%	3.041 (20.74)	3.034 (21.29)	3.063 (20.27)	3.166 (24.12)	3.190 (23.04)	3.913 (21.29)	3.897 (21.29)	3.921 (21.55)	4.028 (24.84)	4.063 (25.21)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	2.477	2.539	2.550	2.584	2.582	3.165	3.180	3.188	3.197	3.206
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	2.642 (6.65)	2.702 (6.43)	2.715 (6.47)	2.729 (5.59)	2.749 (6.49)	3.410 (7.73)	3.424 (7.69)	3.445 (8.05)	3.451 (7.95)	3.455 (7.77)
20%	2.785 (12.44)	2.858 (12.56)	2.759 (8.18)	2.853 (10.39)	2.885 (11.73)	3.566 (12.66)	3.604 (13.35)	3.519 (10.38)	3.585 (12.17)	3.606 (12.47)
30%	2.902 (17.15)	3.011 (18.57)	3.018 (18.32)	3.003 (16.22)	3.074 (19.07)	3.731 (17.89)	3.847 (20.99)	3.817 (19.73)	3.821 (19.55)	3.871 (20.73)
40%	3.258 (31.53)	3.331 (31.19)	3.418 (34.03)	3.494 (35.22)	3.409 (32.03)	4.176 (31.94)	4.256 (33.85)	4.342 (36.21)	4.412 (38.01)	4.327 (34.97)
Imputation using predictive mean matching method										
10%	2.587 (4.45)	2.654 (4.50)	2.652 (3.98)	2.696 (4.34)	2.676 (3.64)	3.342 (5.60)	3.358 (5.60)	3.364 (5.51)	3.399 (6.33)	3.365 (4.96)
20%	2.656 (7.21)	2.736 (7.75)	2.694 (5.61)	2.783 (7.71)	2.791 (8.09)	3.410 (7.75)	3.452 (8.57)	3.432 (7.63)	3.485 (9.02)	3.482 (8.60)
30%	2.709 (9.34)	2.769 (9.06)	2.809 (10.14)	2.759 (6.77)	2.859 (10.75)	3.480 (9.94)	3.541 (11.38)	3.539 (11.01)	3.511 (9.84)	3.591 (12.00)
40%	2.943 (18.80)	2.988 (17.68)	3.024 (18.55)	3.194 (23.61)	3.018 (16.89)	3.782 (19.48)	3.817 (20.06)	3.846 (20.64)	4.026 (25.94)	3.834 (19.59)
Imputation using propensity−score method										
10%	2.611 (5.41)	2.669 (5.10)	2.677 (4.95)	2.696 (4.33)	2.704 (4.73)	3.374 (6.60)	3.378 (6.25)	3.395 (6.50)	3.402 (6.42)	3.403 (6.14)
20%	2.718 (9.73)	2.760 (8.71)	2.701 (5.90)	2.820 (9.12)	2.794 (8.22)	3.489 (10.25)	3.484 (9.58)	3.443 (7.98)	3.536 (10.61)	3.491 (8.87)
30%	2.747 (10.88)	2.802 (10.34)	2.880 (12.91)	2.823 (9.26)	2.942 (13.96)	3.527 (11.44)	3.581 (12.62)	3.630 (13.86)	3.598 (12.56)	3.699 (15.38)
40%	2.981 (20.34)	3.019 (18.89)	3.069 (20.32)	3.177 (22.95)	3.033 (17.46)	3.840 (21.31)	3.873 (21.80)	3.917 (22.87)	4.026 (25.94)	3.872 (20.76)